

Cornell CS 3/5780

Introduction to Machine Learning

Perceptrons

Instructors: Karthik Sridharan and John Thickstun

Quiz 2



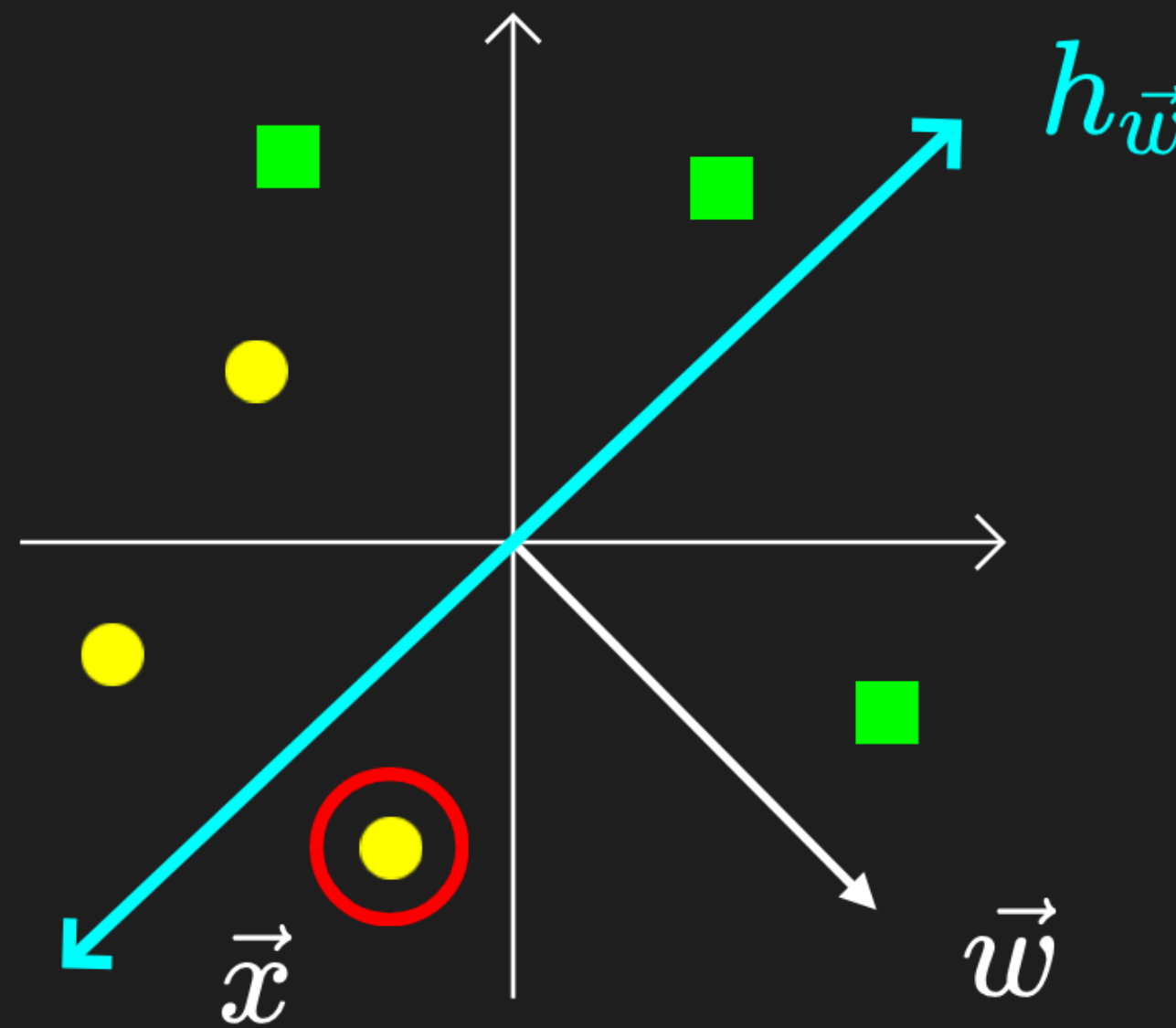
Quiz code is: QUIZ2

Improving a linear classifier

- How can we find a linear classifier?
- Suppose we start with some classifier. What do we do if it misclassifies a data point?

Move **away** from
misclassified
negative points

$$\vec{w} - \vec{x}$$

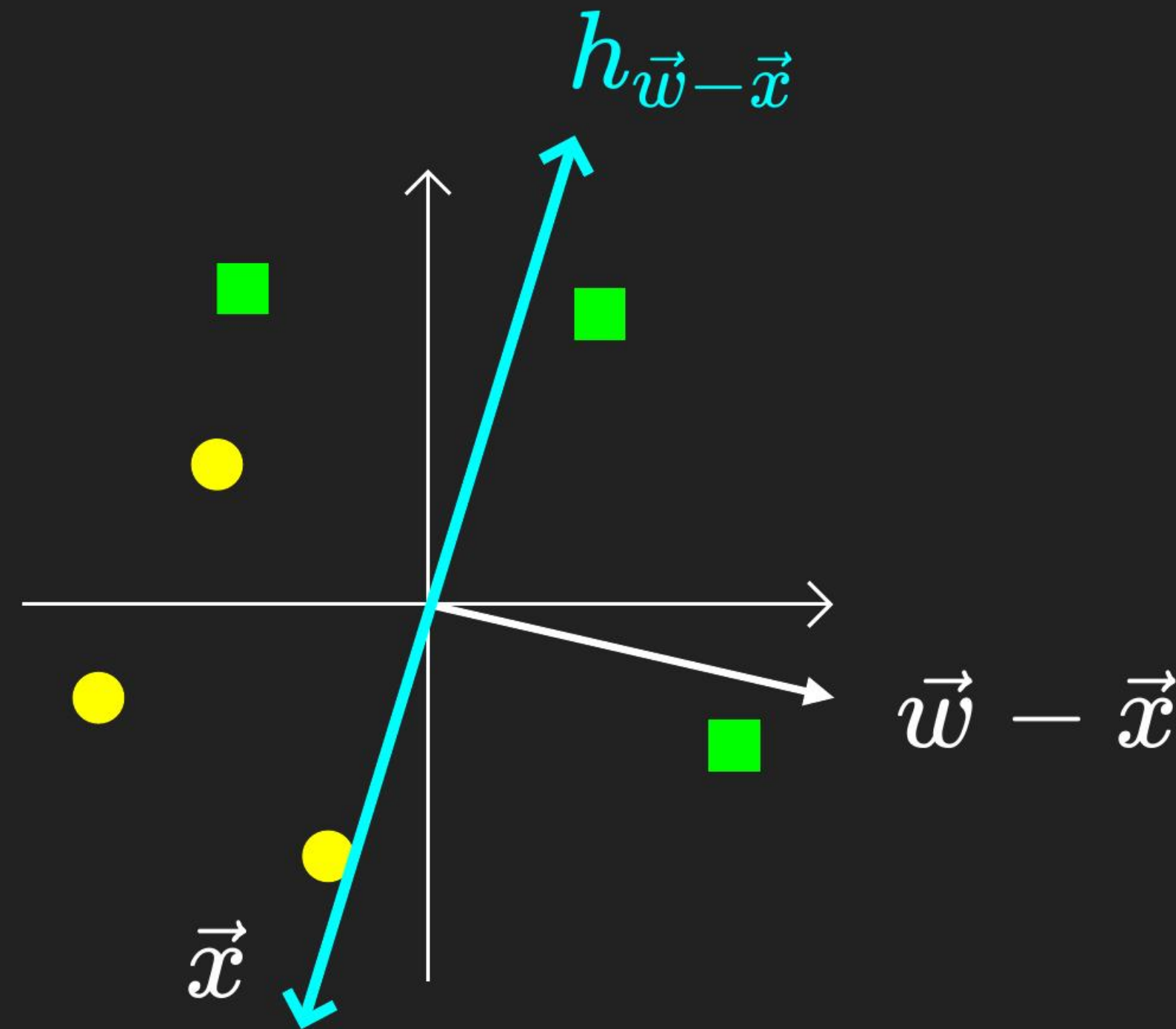


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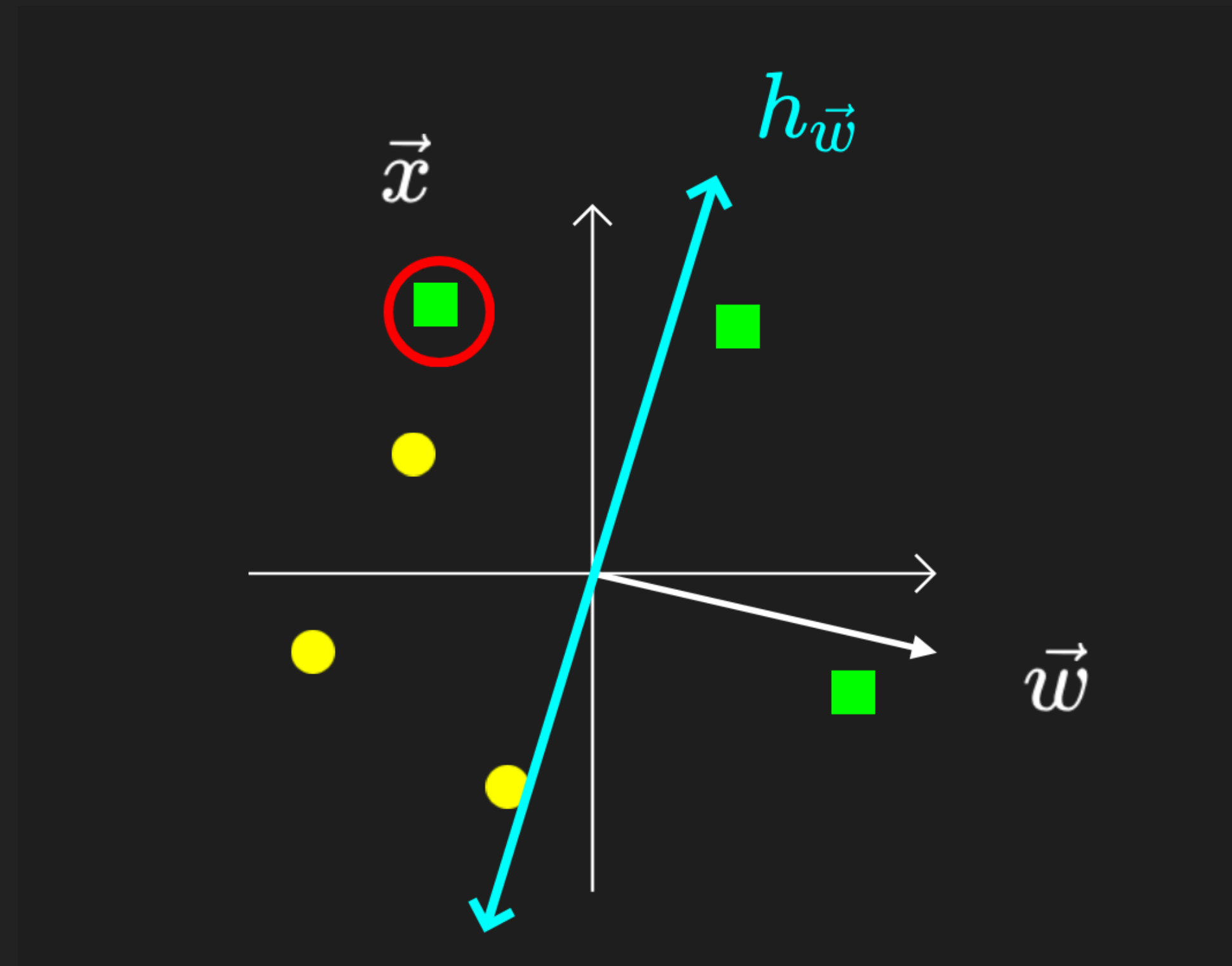


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Move **towards**
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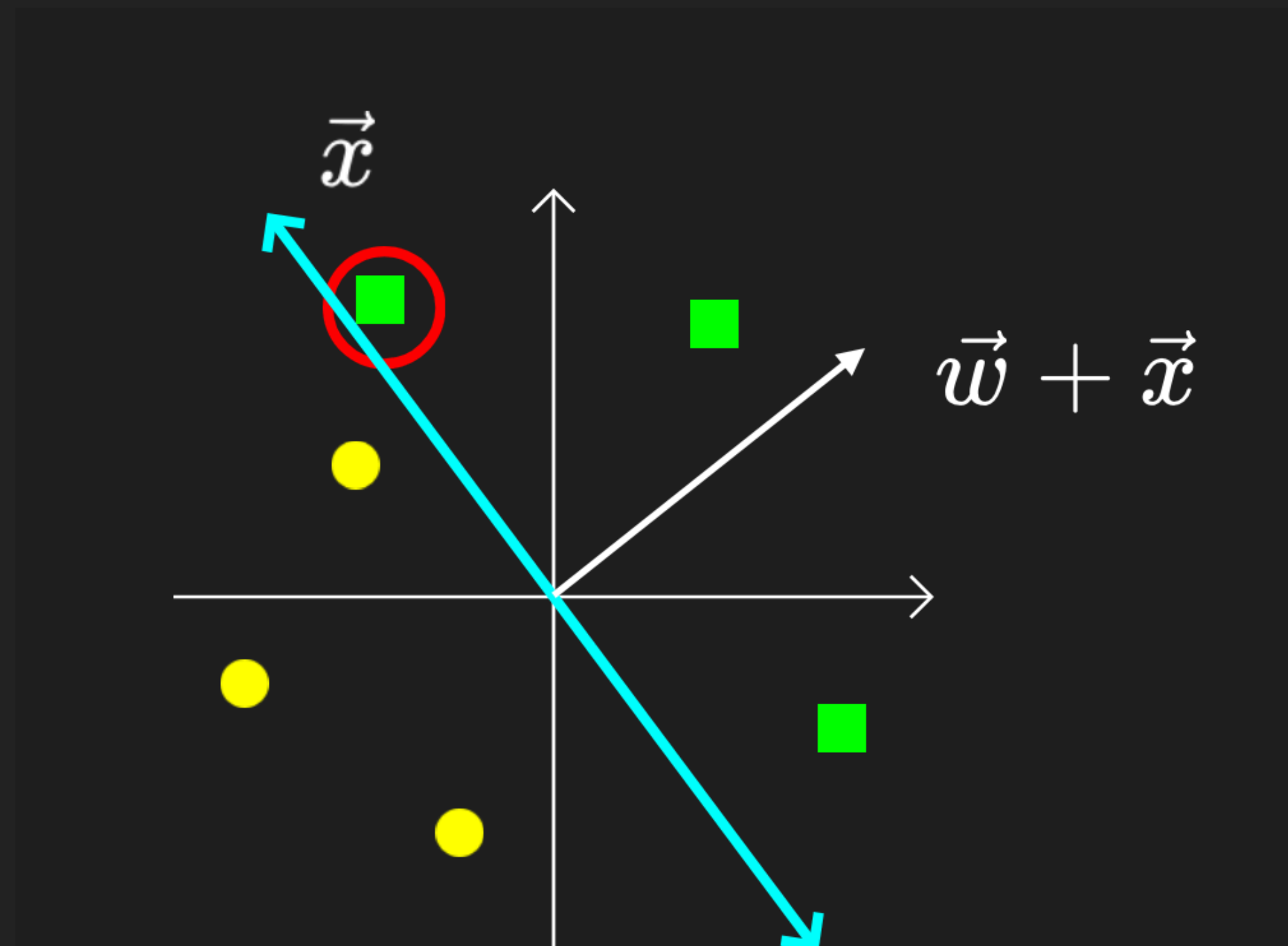
$$\vec{w} + \vec{x}$$

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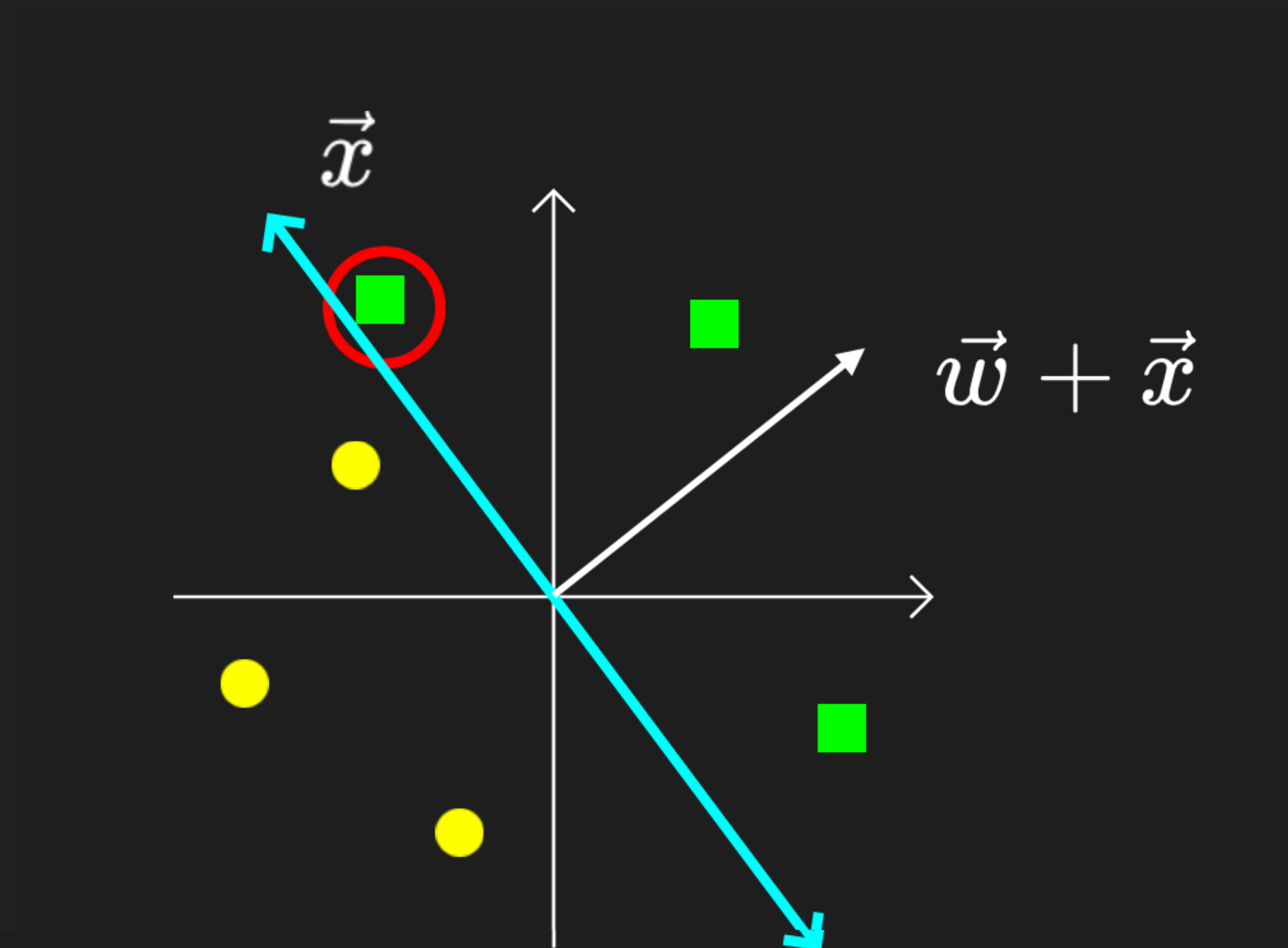
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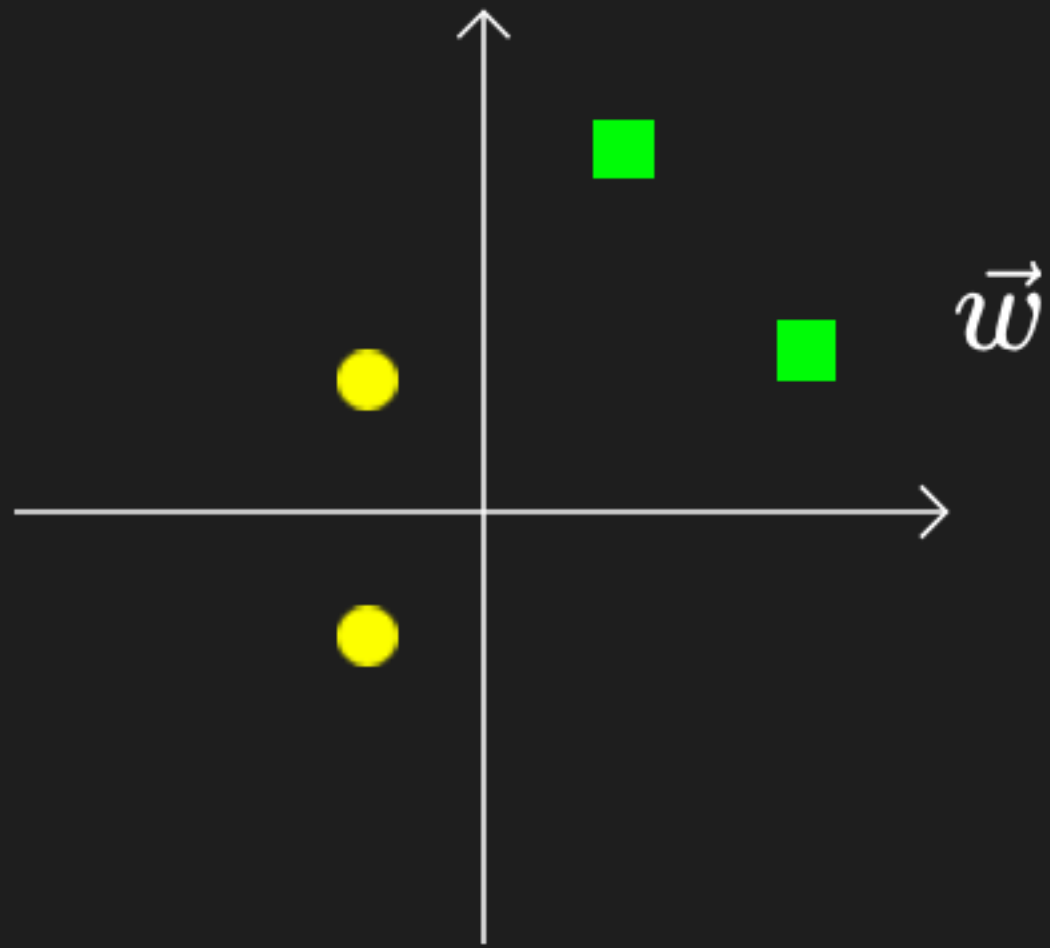
Overall, update
$$\begin{cases} \vec{w} + \vec{x}_i & y_i = 1 \\ \vec{w} - \vec{x}_i & y_i = -1 \end{cases} \iff \vec{w} + y_i \vec{x}_i$$

The Perceptron Algorithm (homogeneous & batch)

- **Input:** training data $\mathcal{S} = \{(\vec{x}_1, y_1), \dots, (\vec{x}_m, y_m)\}$
- **Initialize** $\vec{w}^{(0)} = (0, \dots, 0)$ and $t = 0$
- **While** there is $i \in [m]$ such that $y_i(\vec{w}^{(t)} \cdot \vec{x}_i) \leq 0$ then,
 - $\vec{w}^{(t+1)} = \vec{w}^{(t)} + y_i \vec{x}_i$
 - $t \leftarrow t + 1$
- **End While**
- **Output** $\vec{w}^{(t)}$

Interactive Demo: <https://mlweb.loria.fr/book/en/perceptron.html>

Perceptron Example



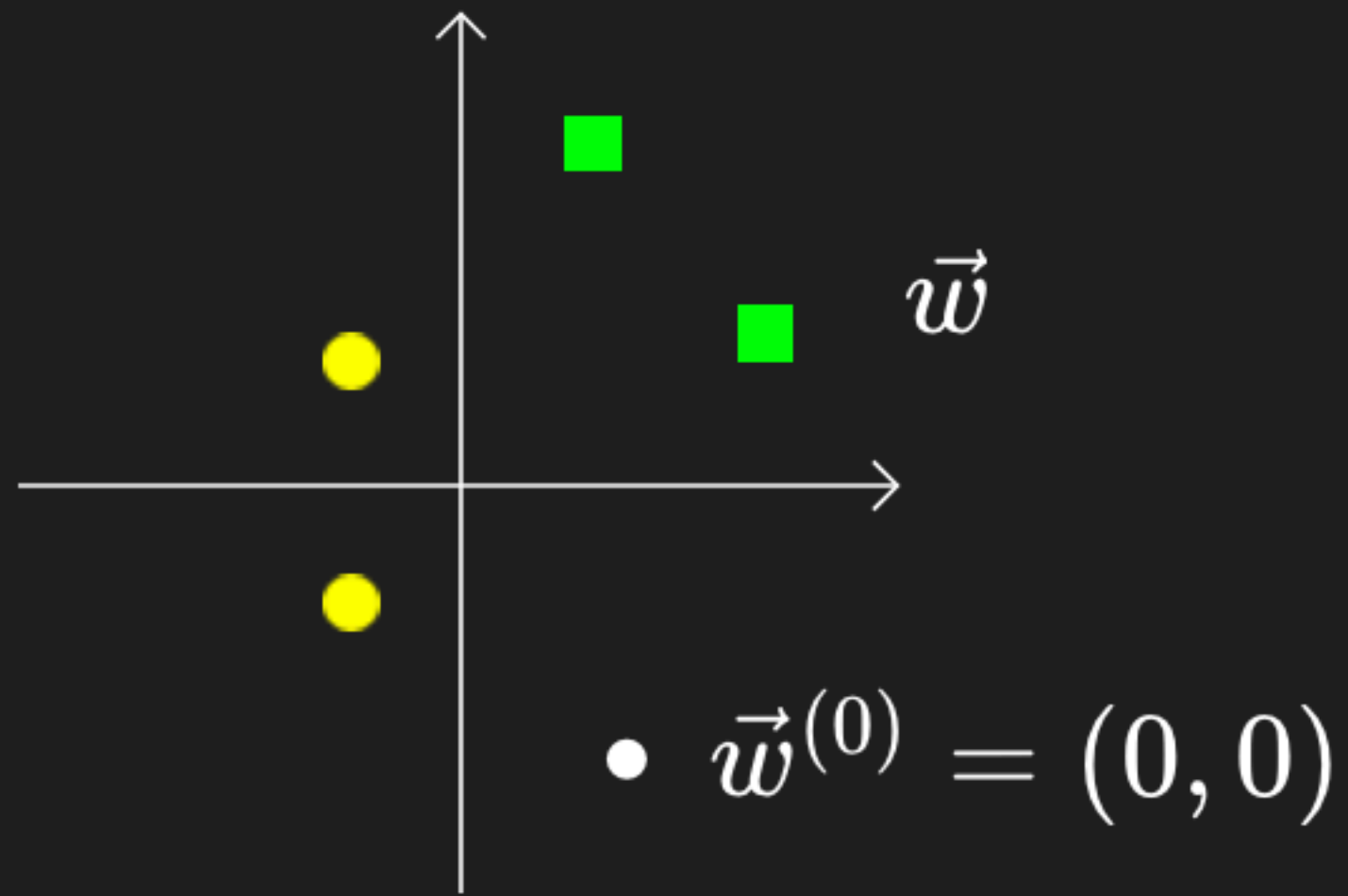
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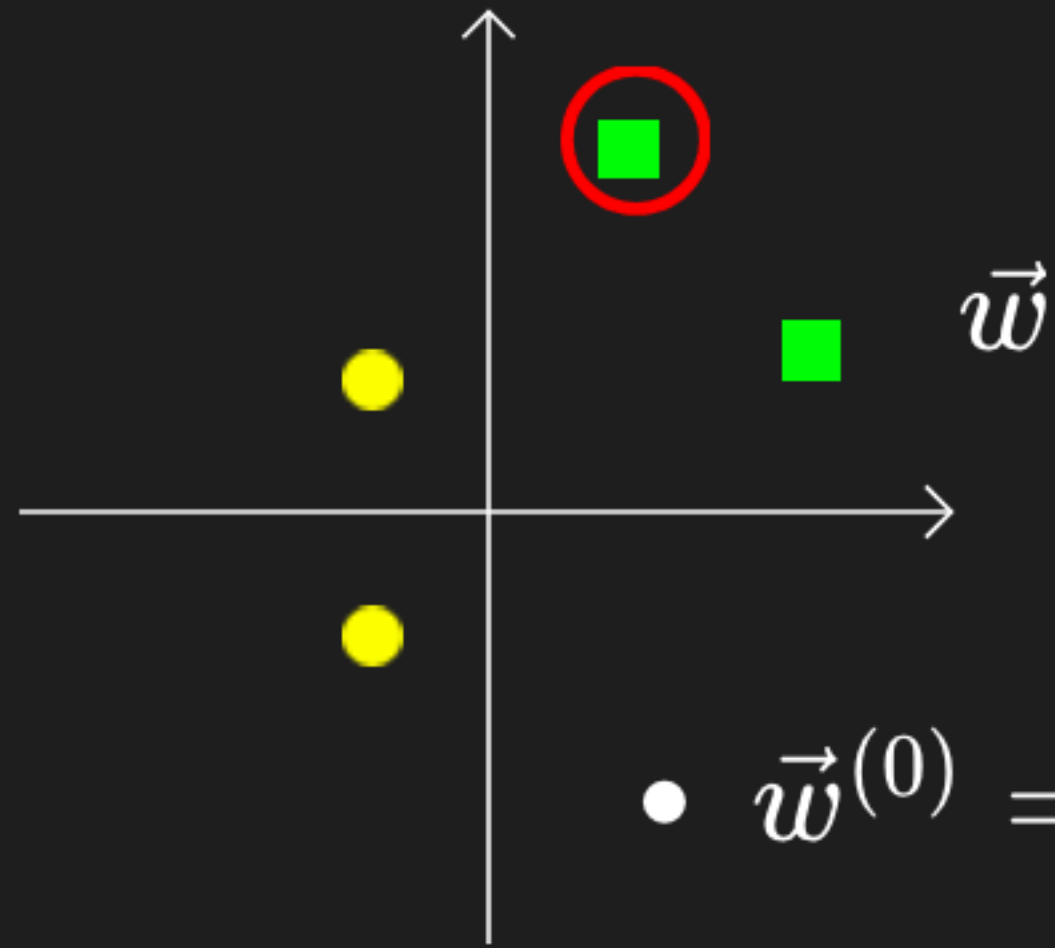
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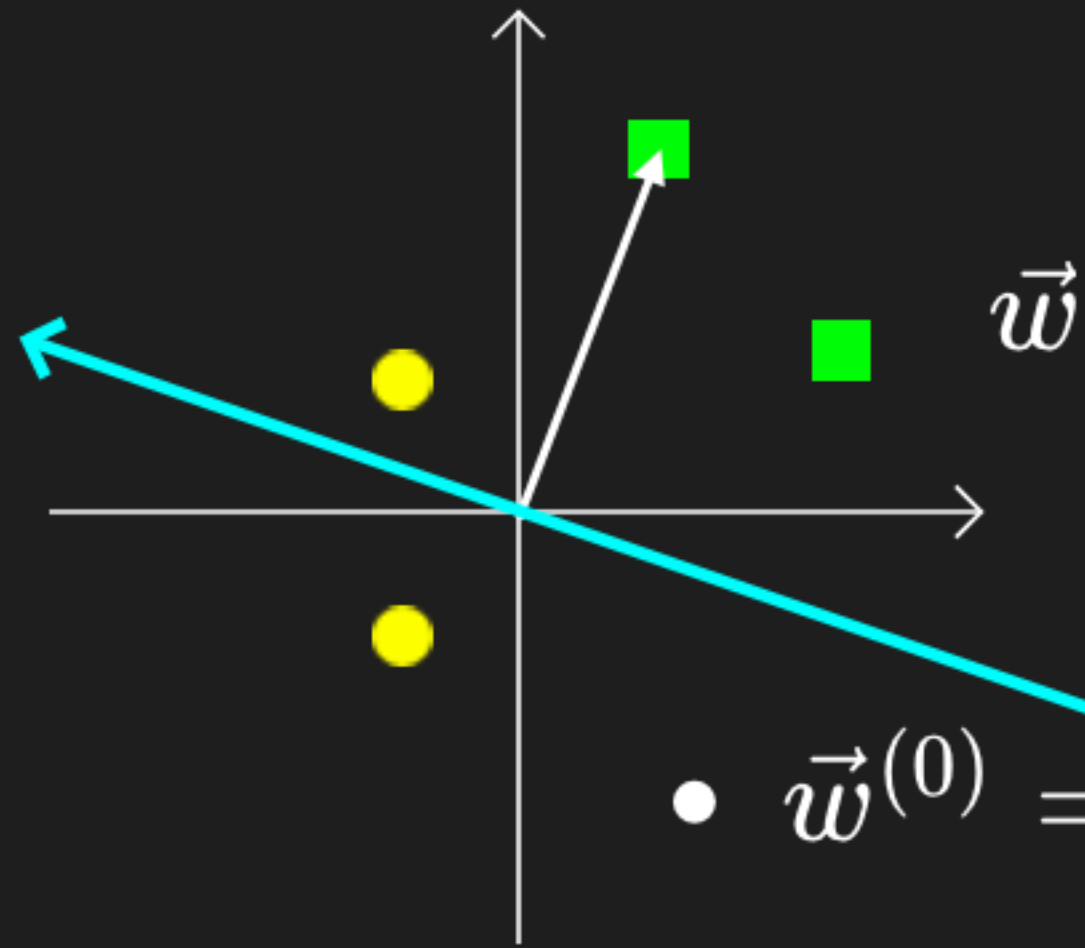
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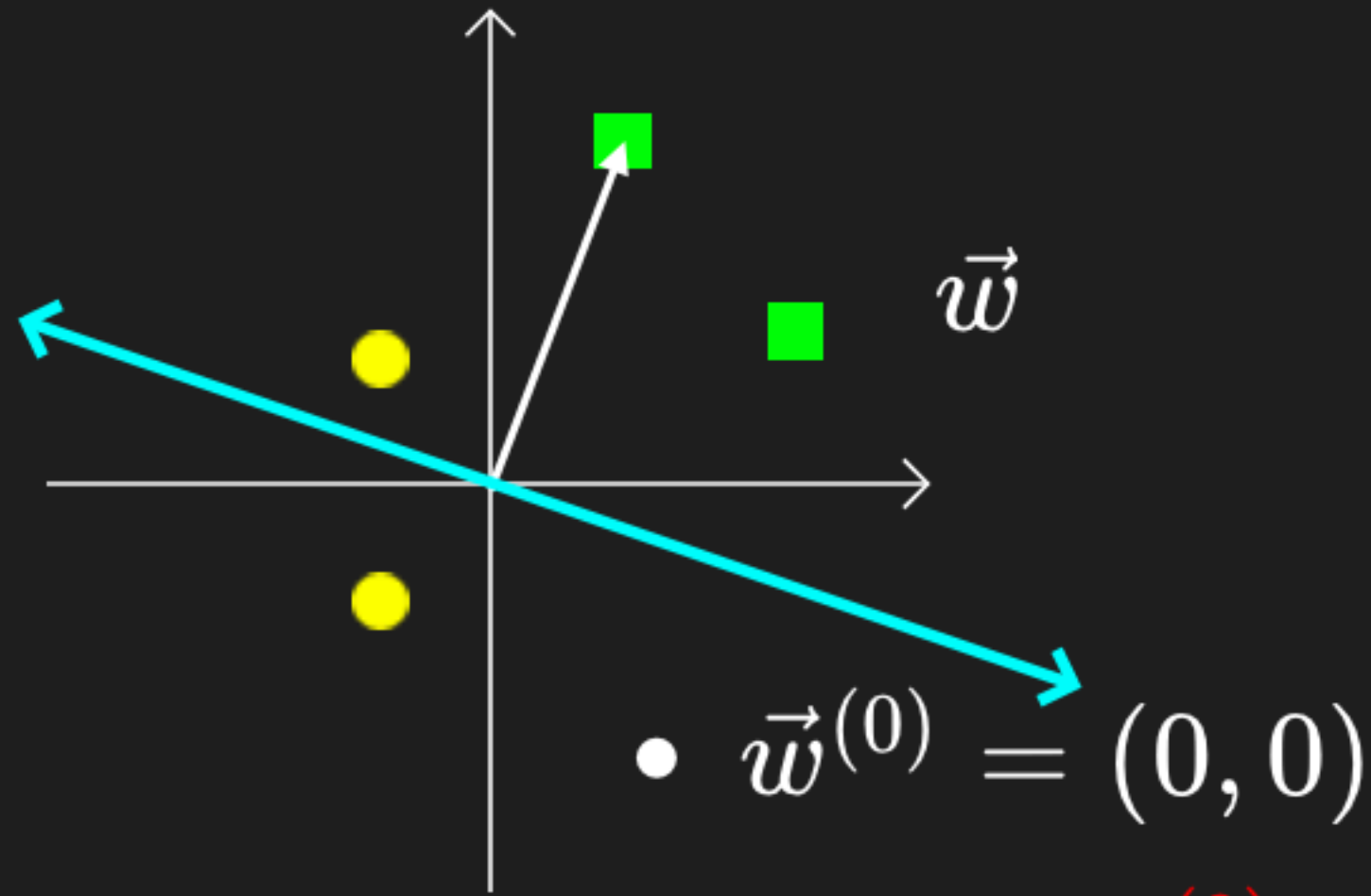
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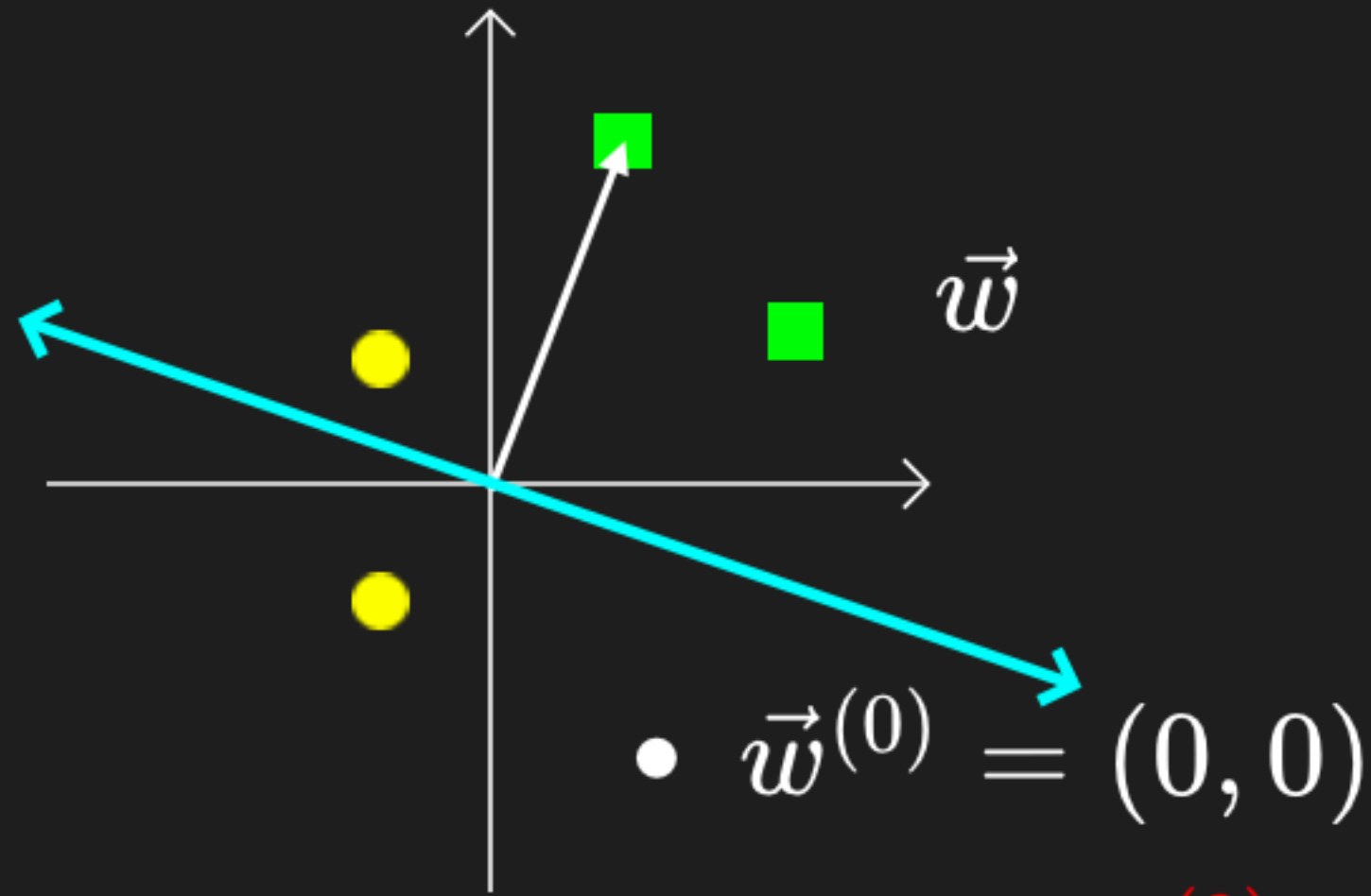
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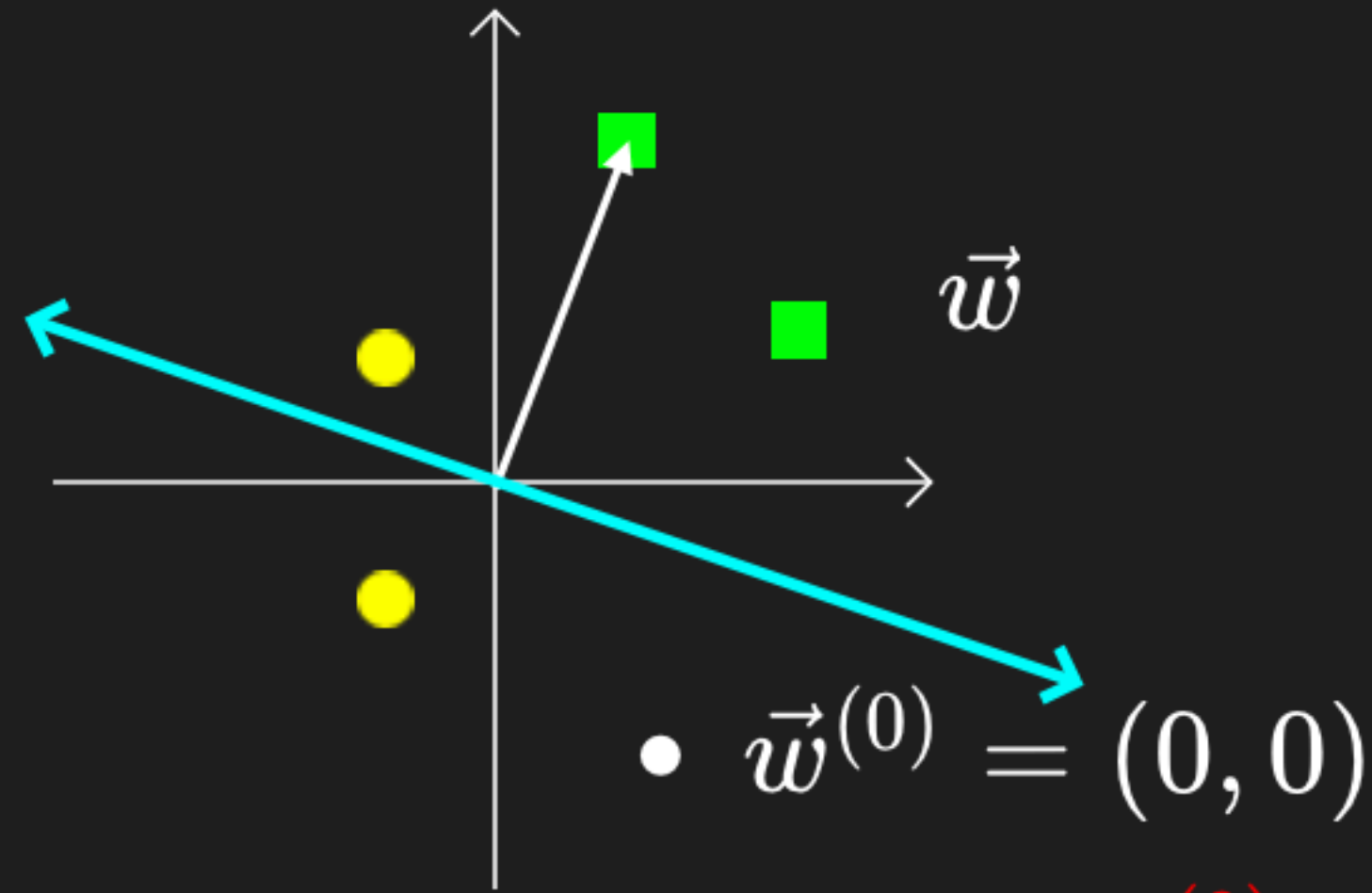
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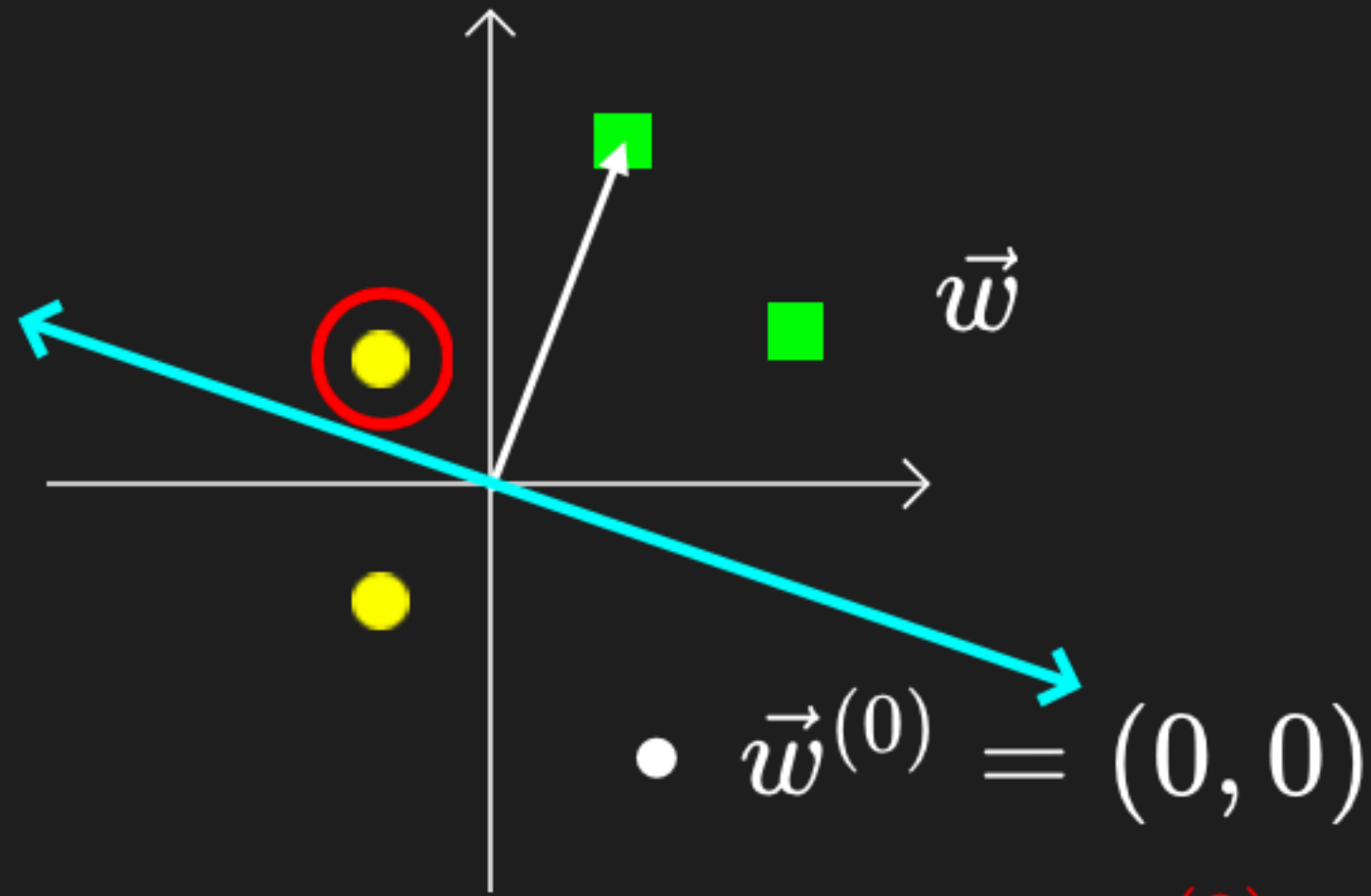
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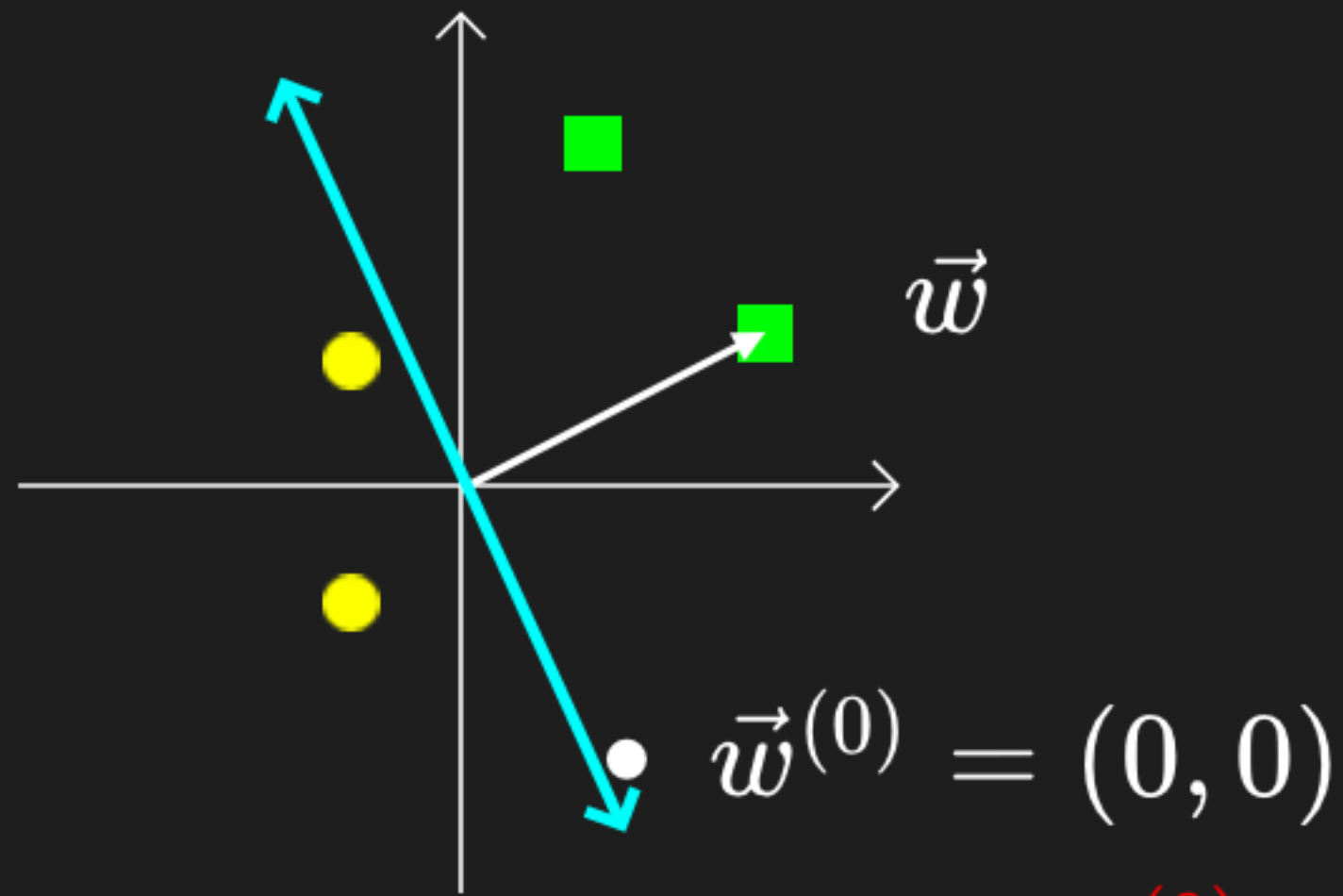
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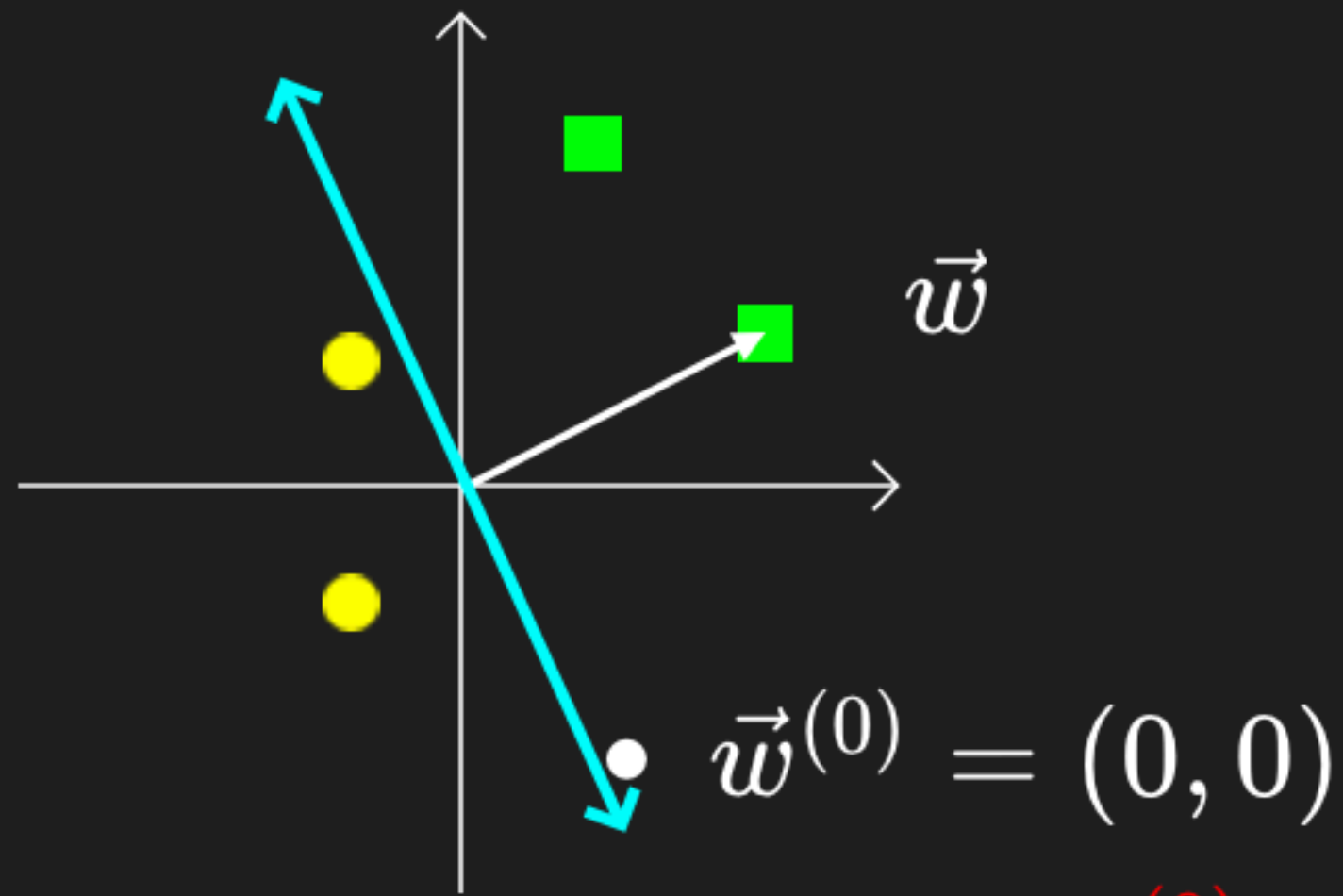


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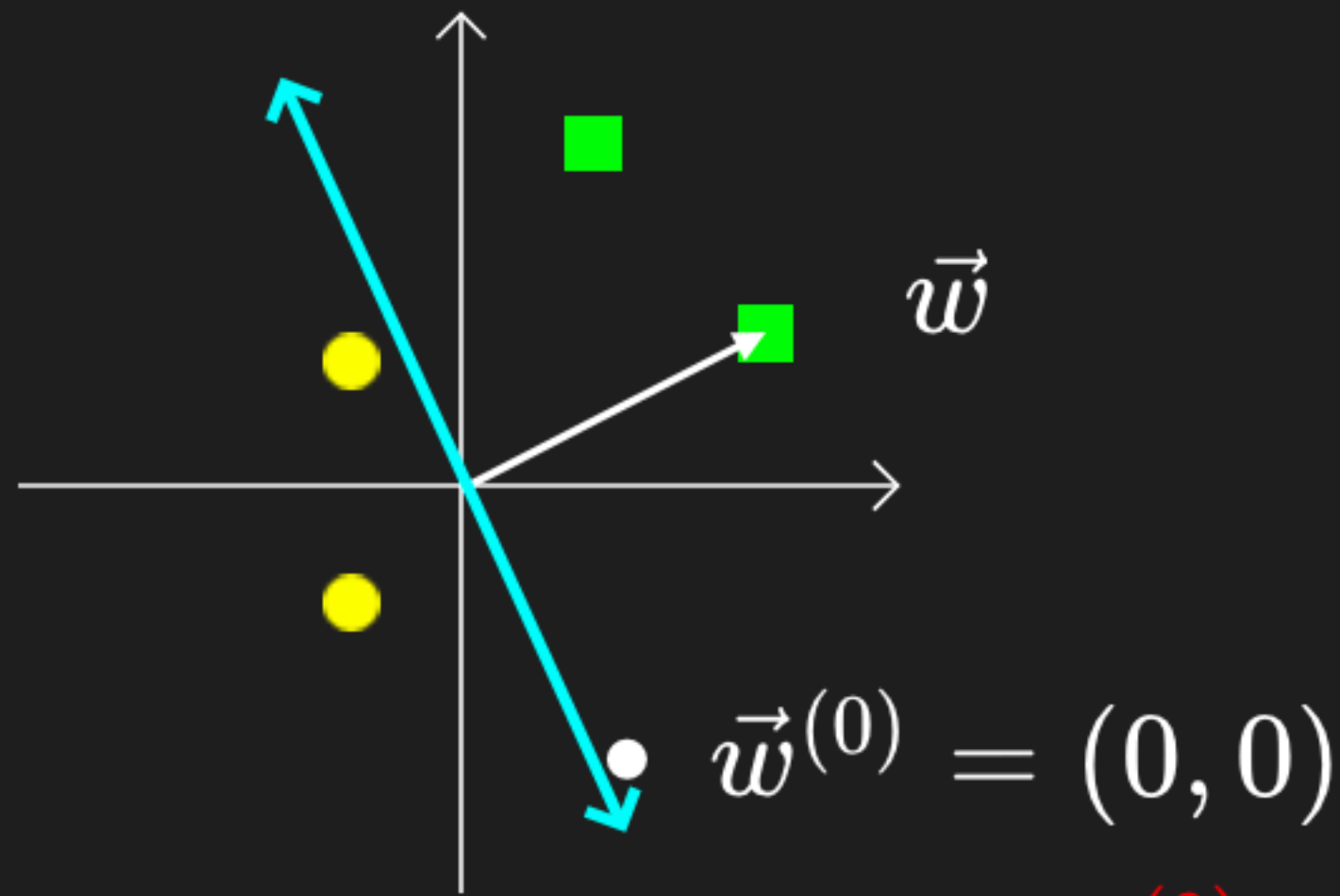


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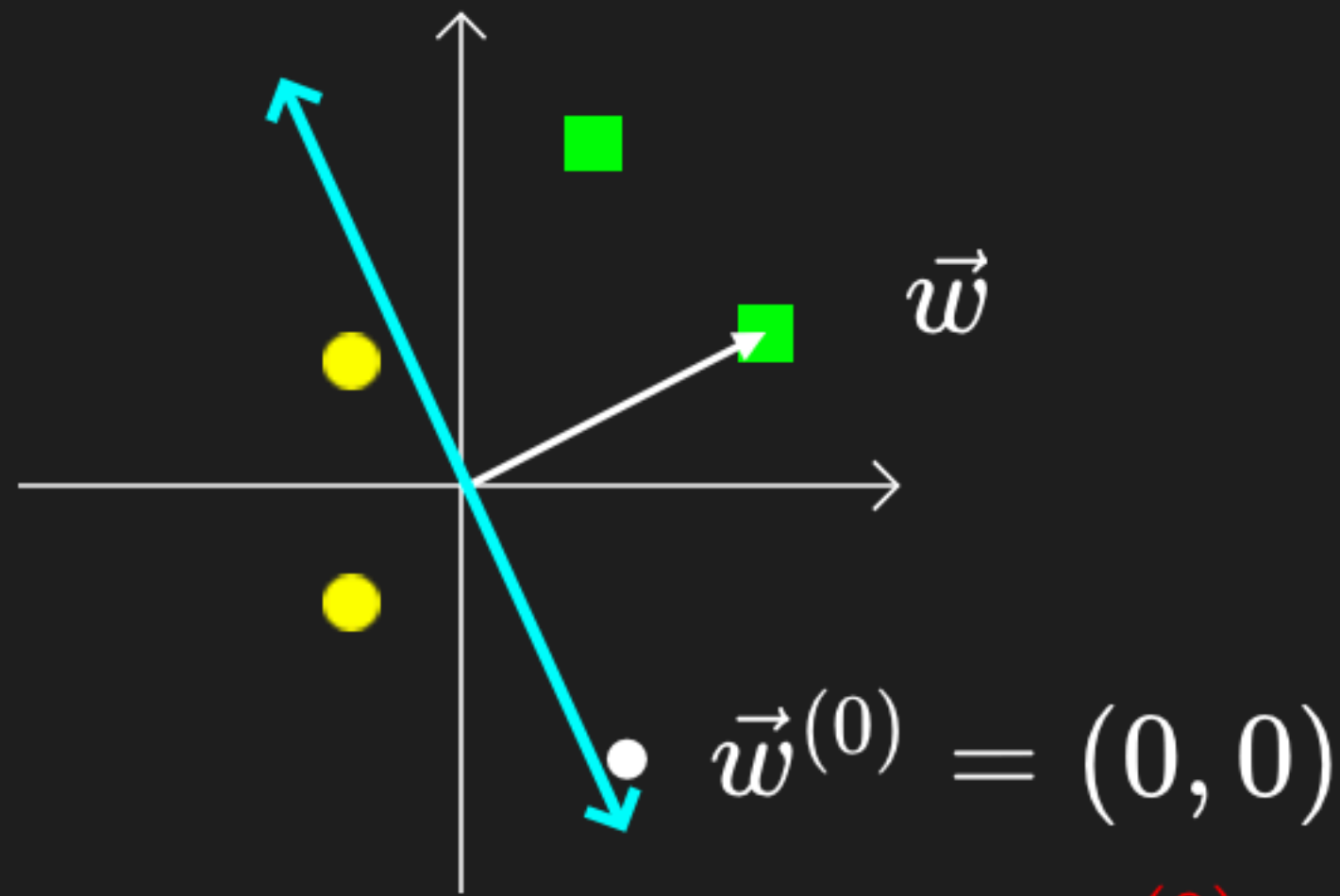
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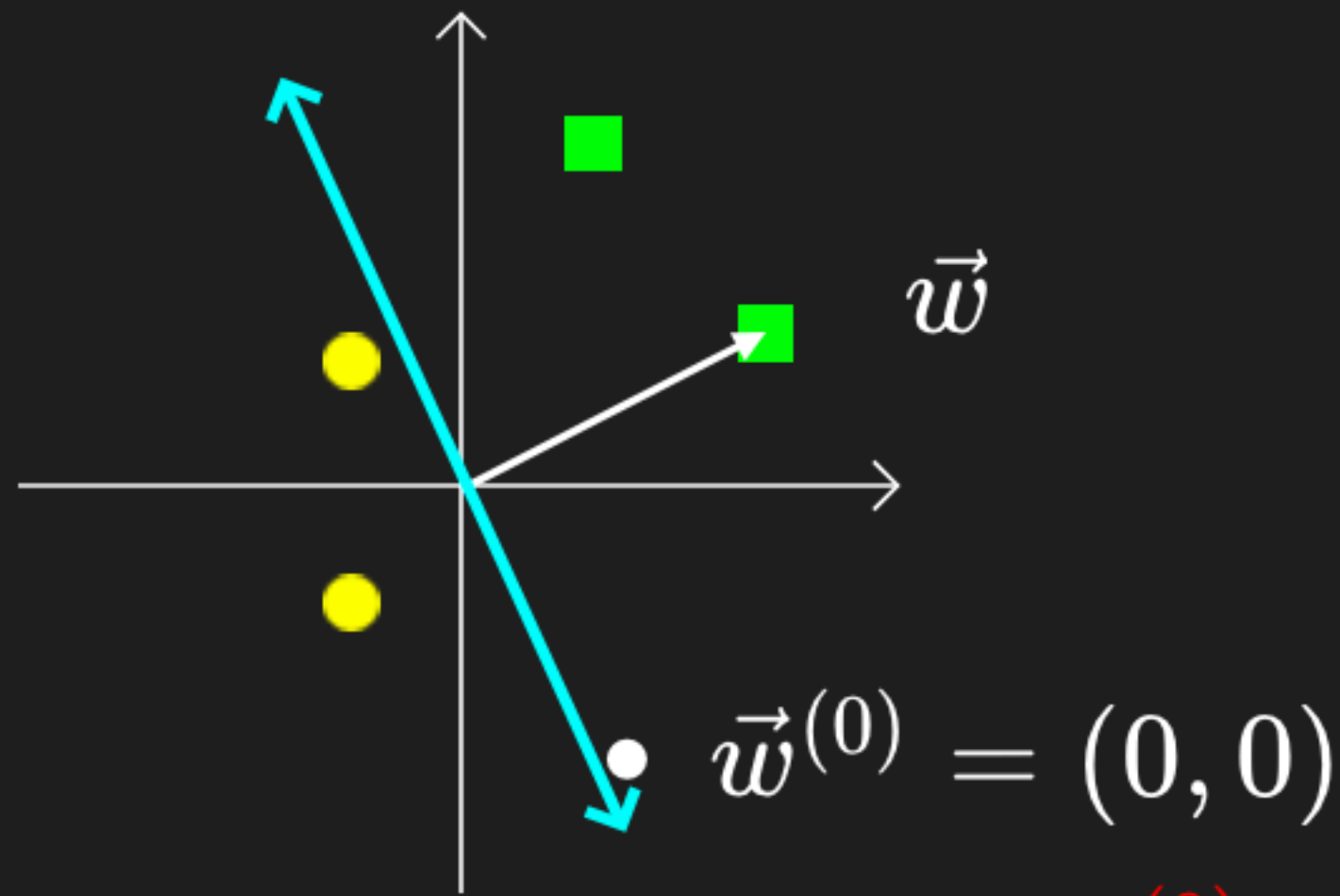
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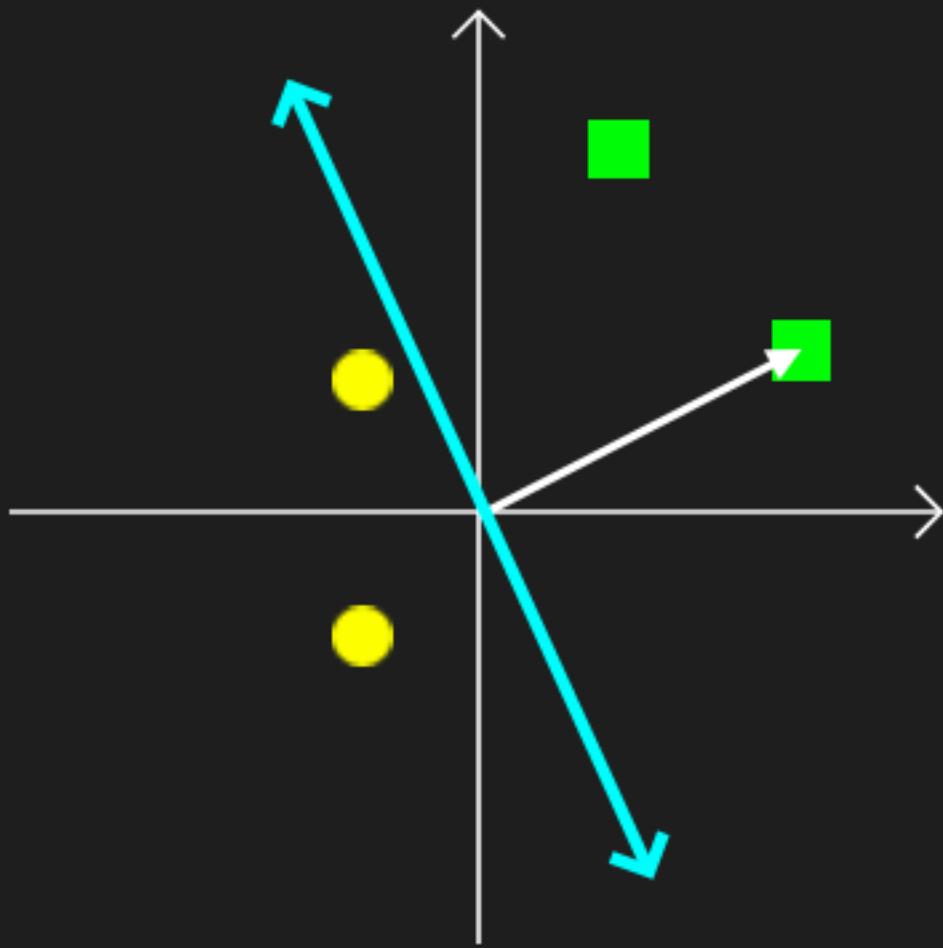
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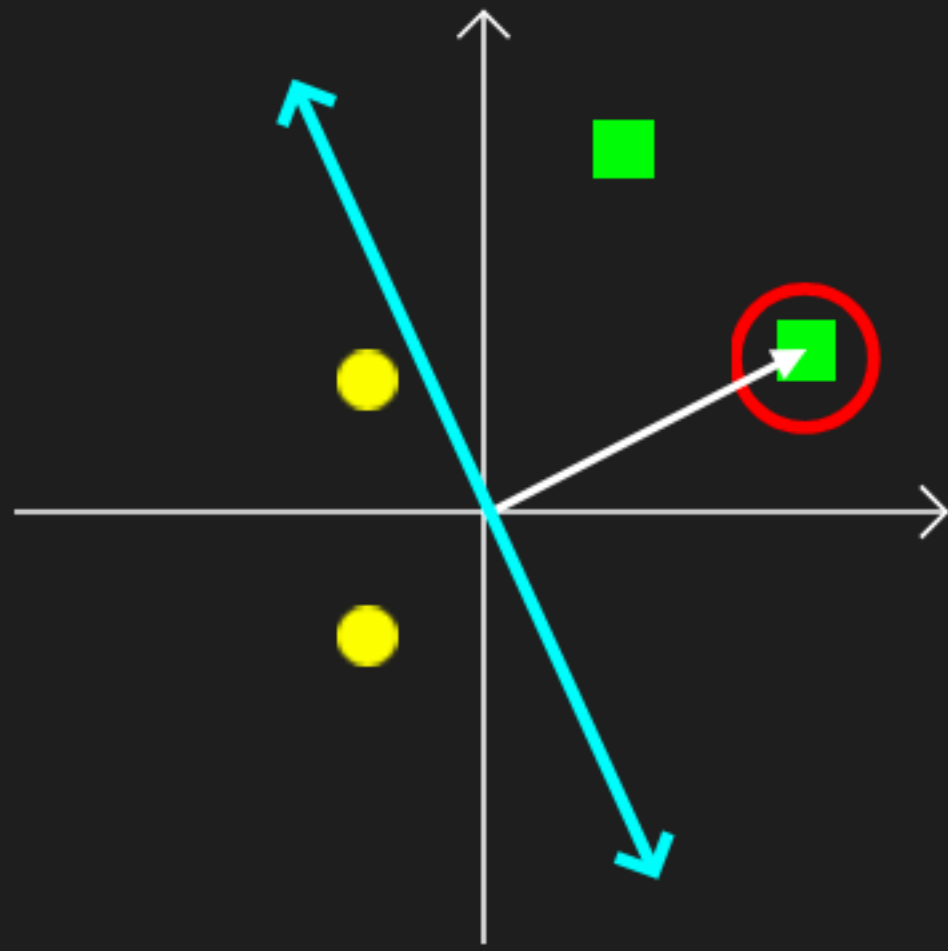
Changing the data order



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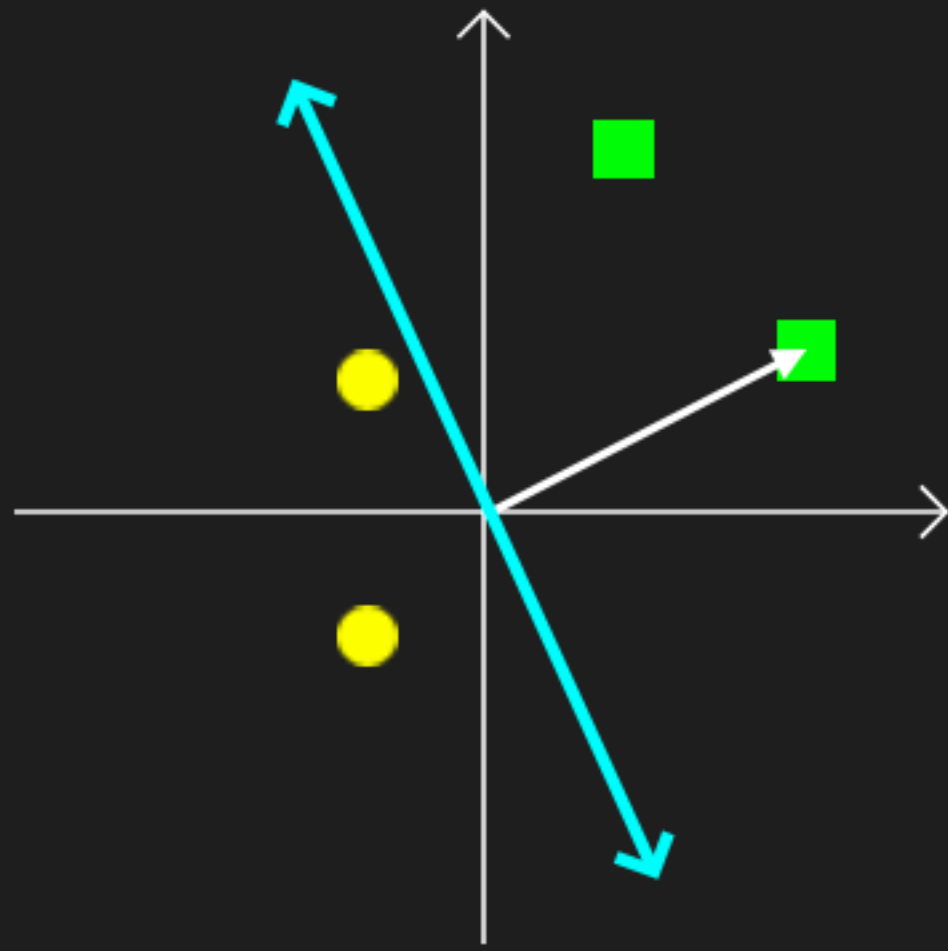


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Changing the data order



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**faster
convergence**

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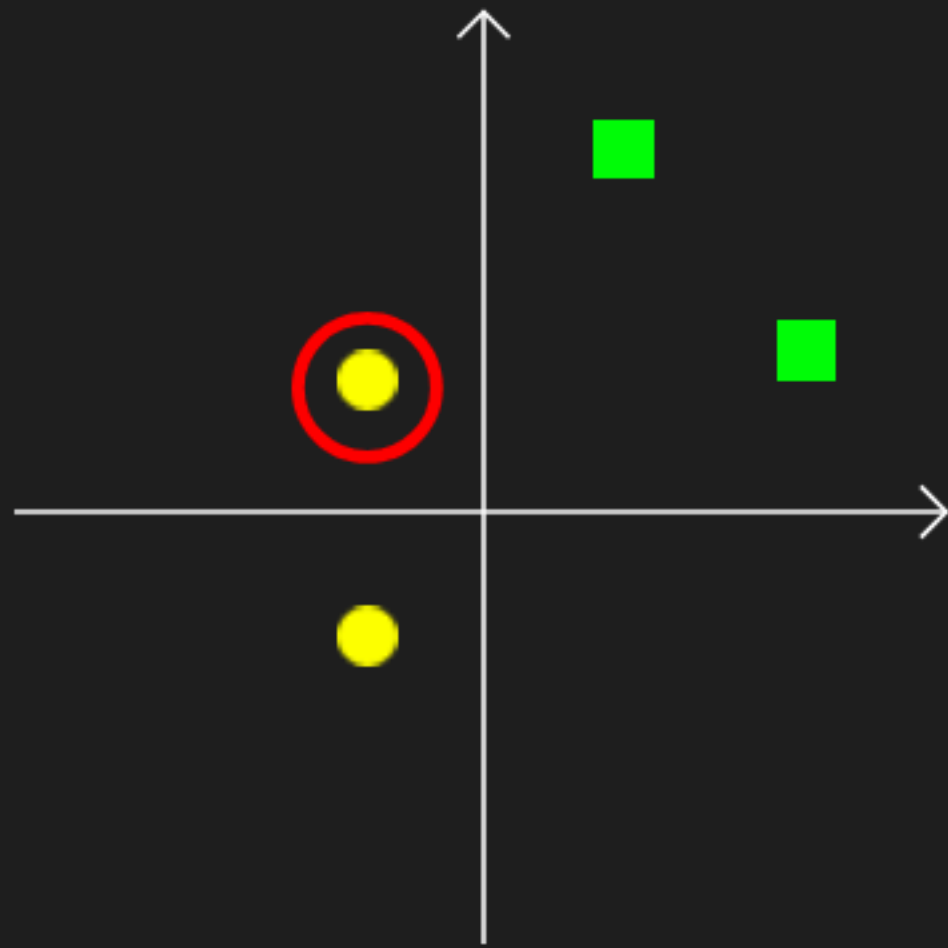
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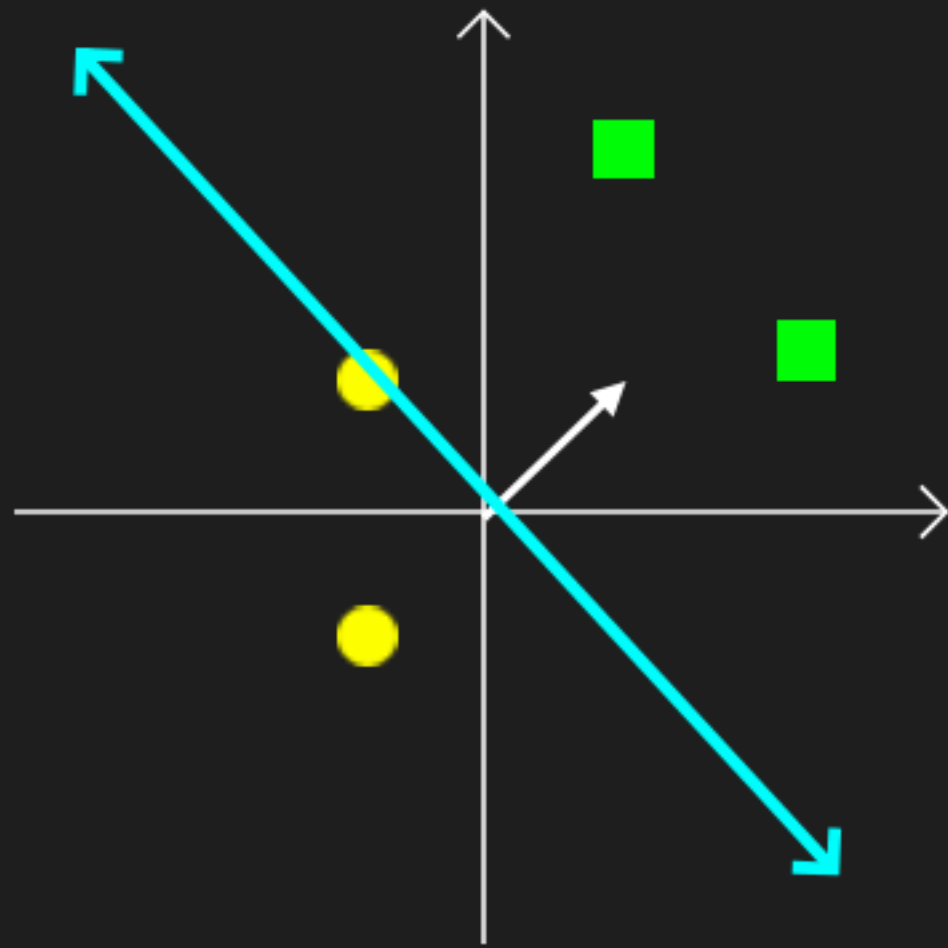
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- Update on (x_3, y_3)

Changing the data order



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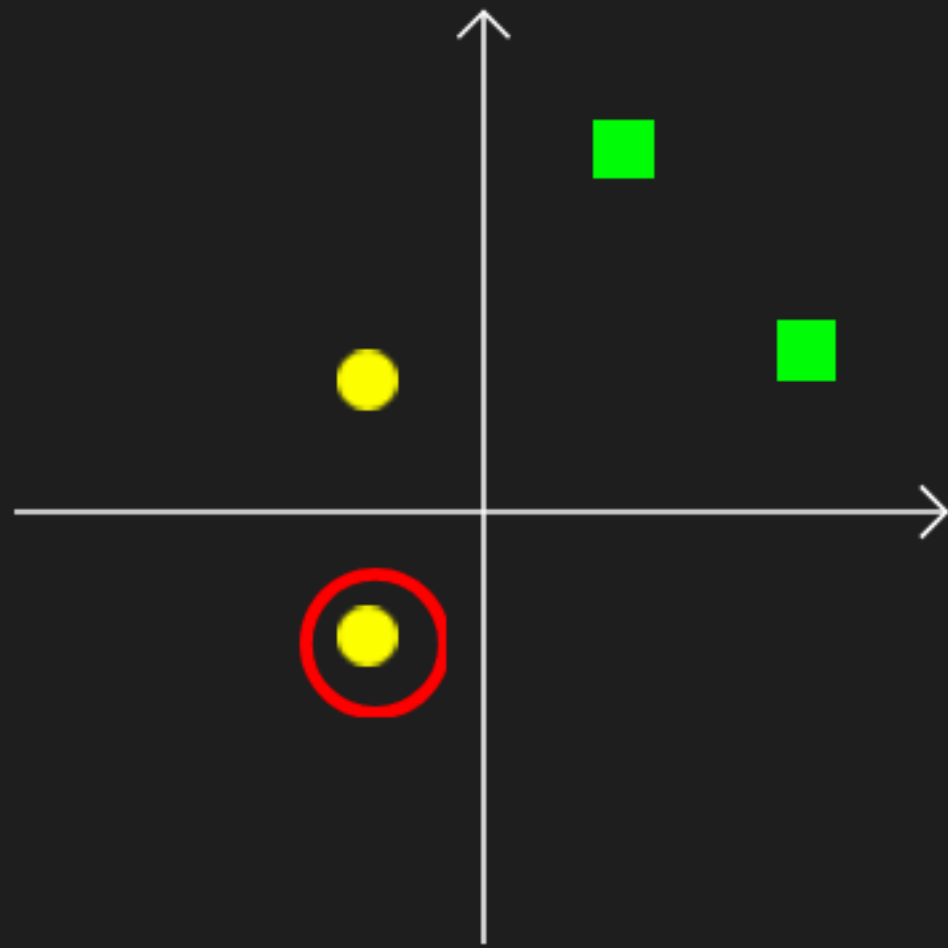
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- Update on (x_3, y_3)

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Changing the data order



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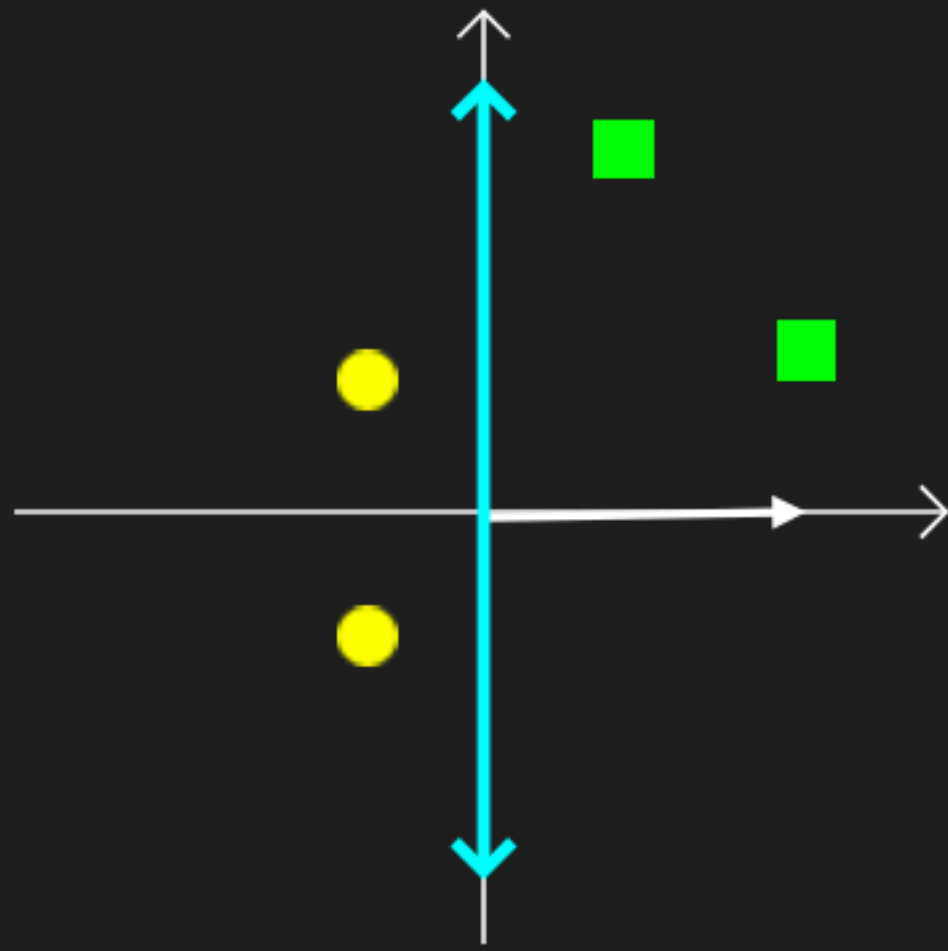
- $\vec{w}^{(2)} = \vec{w}^{(1)} - \vec{x}_4 = (2, 1)$

- Update on (x_3, y_3)

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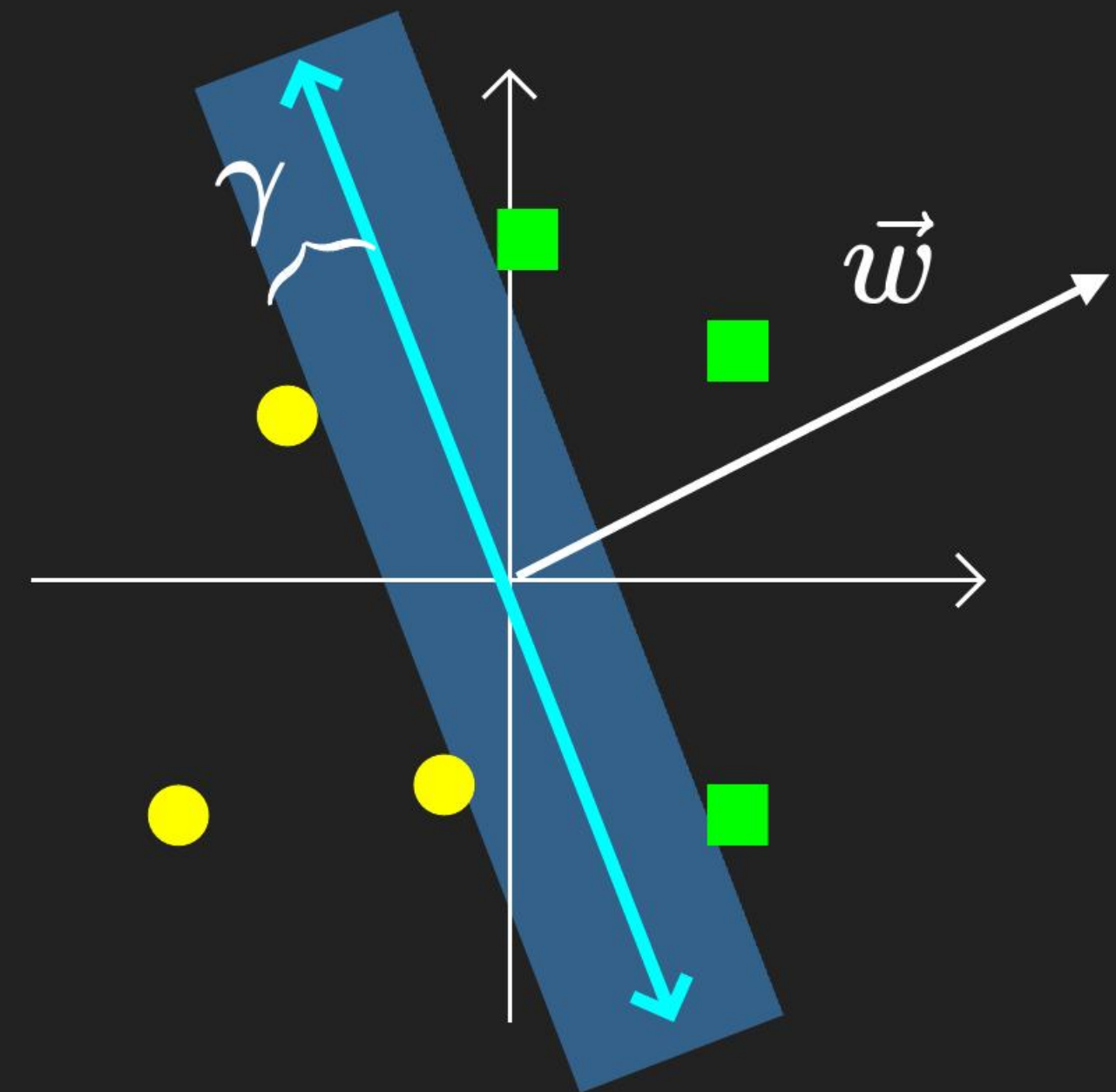
- Update on (x_4, y_4)

- $\vec{w}^{(1)} = (2, 0)$

different classifier

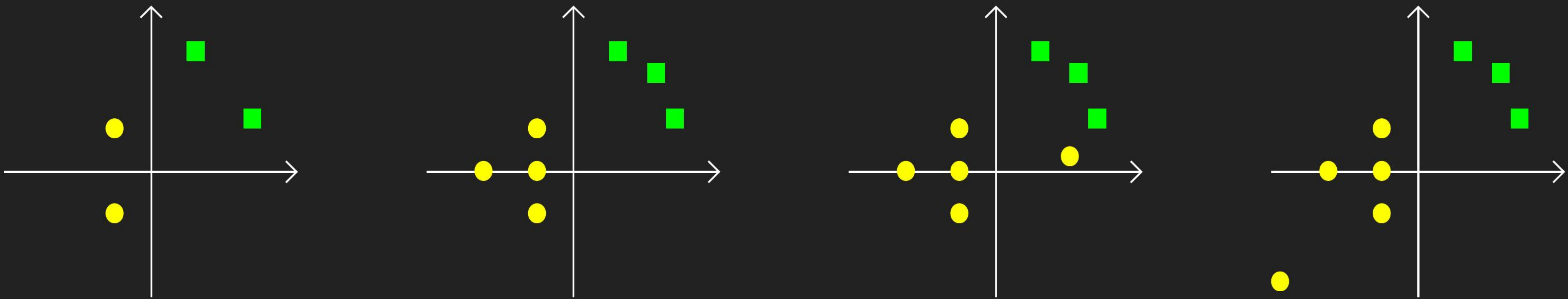
Margin

- A data set $\mathcal{S}$ is linearly separable with a **(geometric) margin** γ if
 - there is a linear classifier $h_{\vec{w}}$ that is consistent with $\mathcal{S}$
 - and the distance of any instance $\mathcal{S}$ to the decision boundary of $h_{\vec{w}}$ is at least γ
- Mathematical definition: There is $\vec{w}$ such that $\|\vec{w}\| = 1$ and for all data points $(\vec{x}_i, y_i) \in \mathcal{S}$
$$\begin{cases} \vec{w} \cdot \vec{x}_i \geq \gamma & y_i = 1 \\ \vec{w} \cdot \vec{x}_i \leq -\gamma & y_i = -1 \end{cases} \iff y_i (\vec{w} \cdot \vec{x}_i) \geq \gamma$$
- Larger margin $\implies$ Easier to find consistent linear classifier



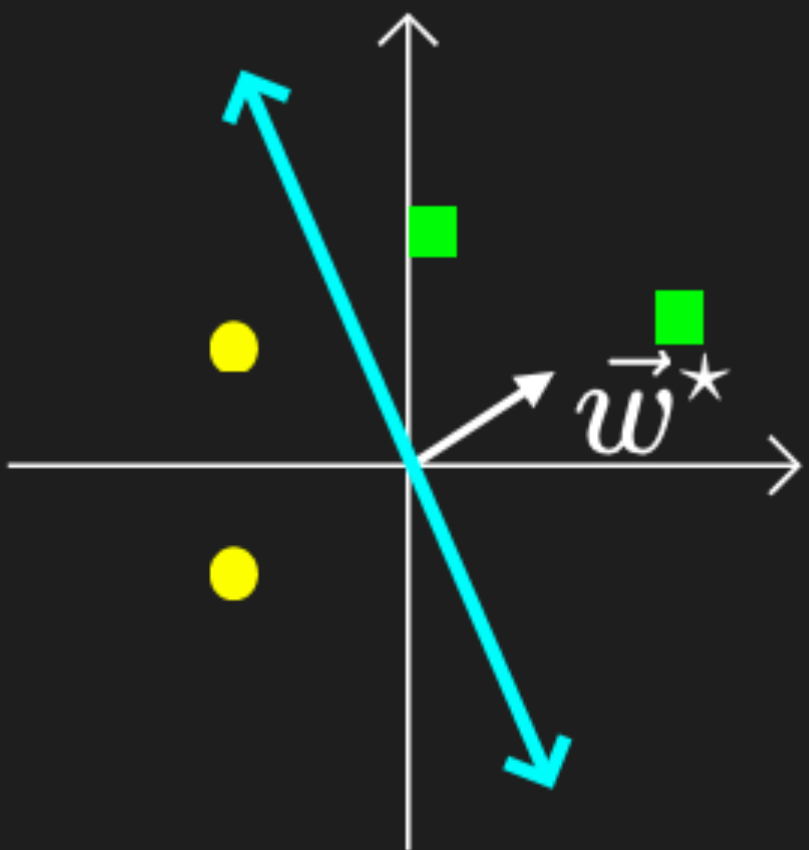
Convergence of Perceptron

- **Theorem:** Given a dataset $\mathcal{S} = \{(\vec{x}_1, y_1), \dots, (\vec{x}_m, y_m)\}$ and a radius R such that $\|\vec{x}_i\| \leq R$ for all $i \in [m]$. If $\mathcal{S}$ is linearly separable with (geometric) **margin** γ , then Perceptron makes at most R^2/γ^2 updates before finding a consistent linear classifier.
- *As long as the data is linearly separable with a margin, then the Perceptron algorithm terminates after finitely many updates*



Proof Idea

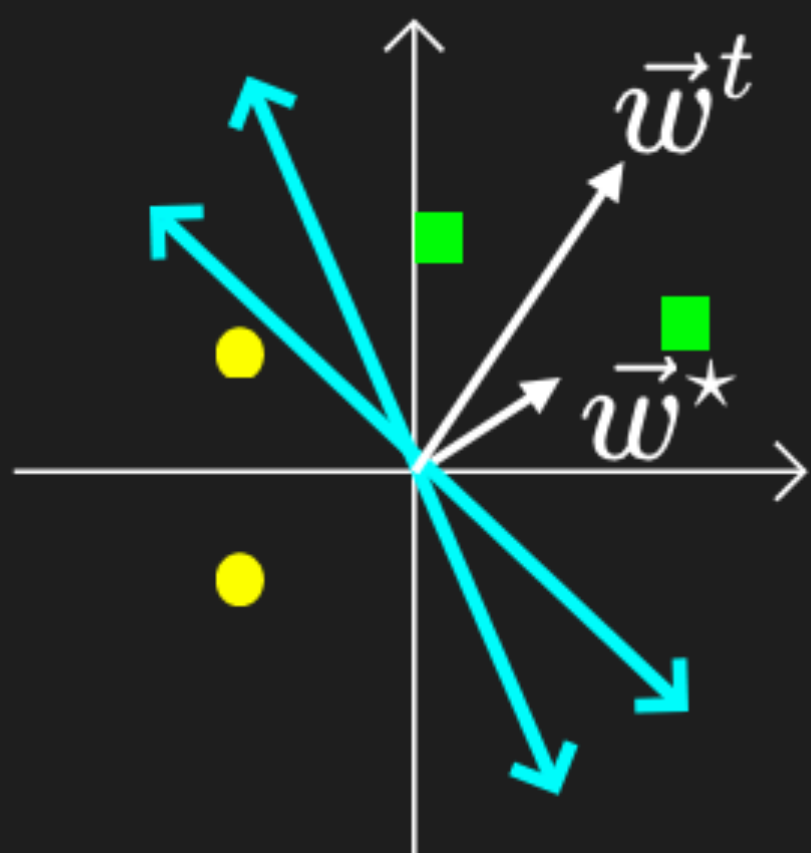
- Since $\mathcal{S}$ is linearly separable with **margin** γ , there is a $h_{\vec{w}^*}$ with γ margin
 - this means $\|\vec{w}^*\|_2 = 1$ and $y_i(\vec{w}^* \cdot \vec{x}_i) \geq \gamma$ for all $i \in [m]$



Proof Idea

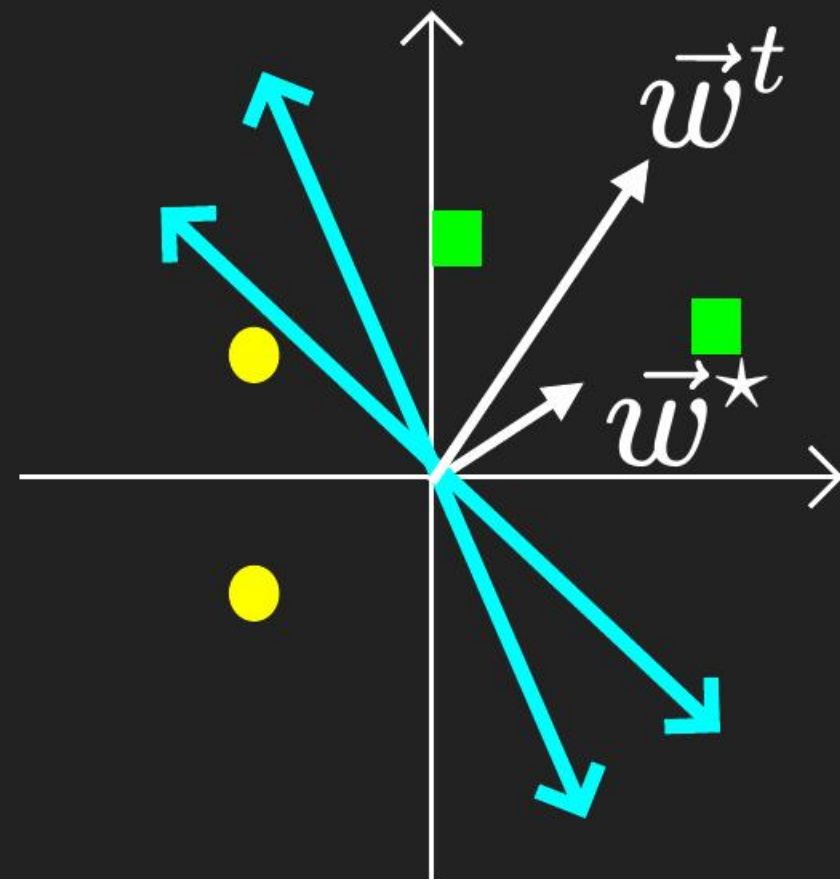
- Since $\mathcal{S}$ is linearly separable with **margin** γ , there is a $\vec{w}^*$ with γ margin
 - this means $\|\vec{w}^*\|_2 = 1$ and $y_i(\vec{w}^* \cdot \vec{x}_i) \geq \gamma$ for all $i \in [m]$
- We will show that within $t = R^2/\gamma^2$ steps, $\vec{w}^{(t)}$ is close to $\vec{w}^*$ in angle
- Recall that the angle is given by

$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$



Proof Outline

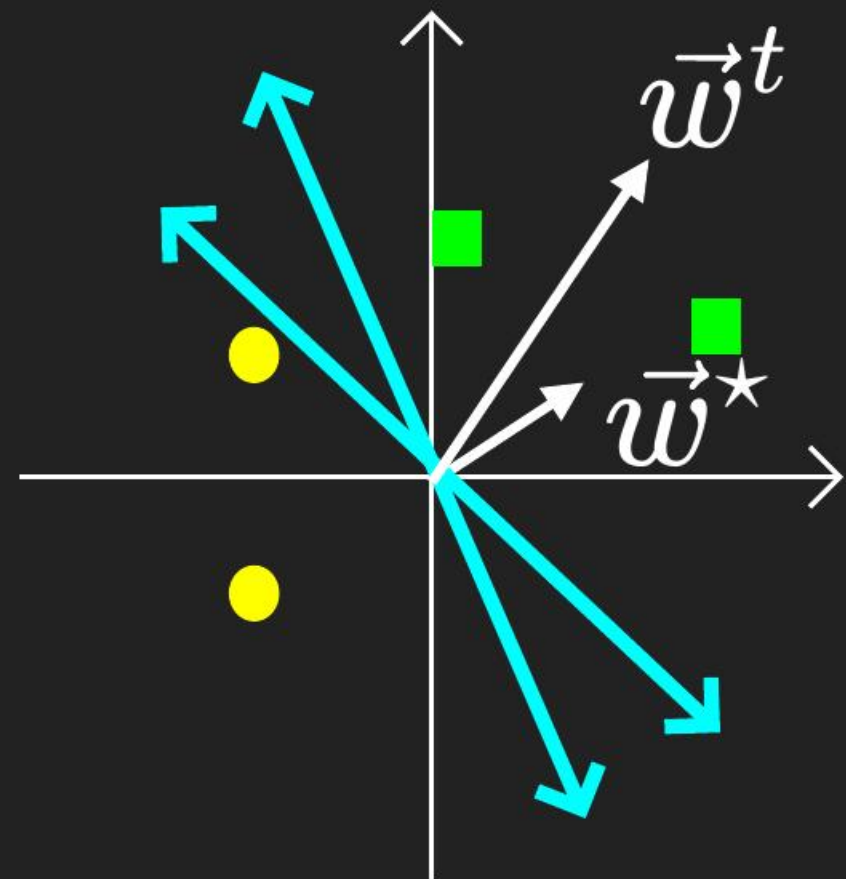
- Goal: Prove that Perceptron terminates after at most R^2/γ^2 steps
- Step 1: Show that $\vec{w}^* \cdot \vec{w}^{(t)}$ is large (find a lower bound)



$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$

Proof Outline

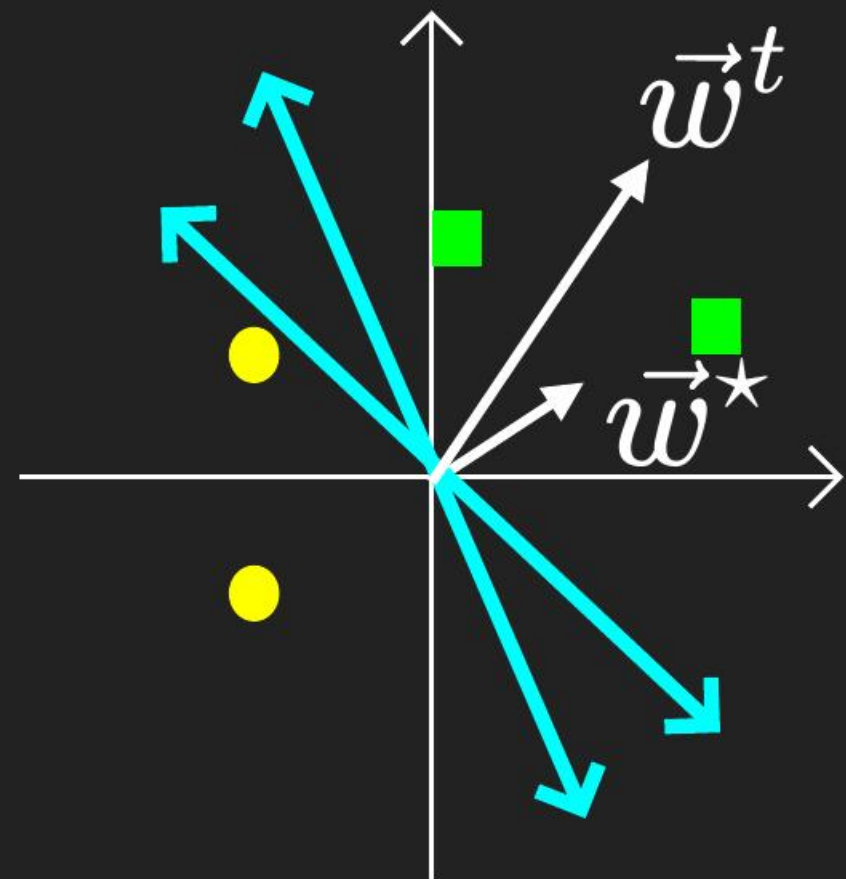
- Goal: Prove that Perceptron terminates after at most R^2/γ^2 steps
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- Step 2: Show that $\|\vec{w}^{(t)}\|$ is not too large (find an upper bound)



$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$

Proof Outline

- Goal: Prove that Perceptron terminates after at most R^2/γ^2 steps
- Step 1: Show that $\vec{w}^* \cdot \vec{w}^{(t)}$ is large (find a lower bound)
- Step 2: Show that $\|\vec{w}^{(t)}\|$ is not too large (find an upper bound)
- Step 3: combine 1 & 2 to show that if there are more than R^2/γ^2 updates, then the cosine becomes larger than one!
 - Proof by contradiction



$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$

Step 1: $\vec{w}^\star \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^\star \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^\star \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^\star \cdot \vec{w}^{(t-1)}$ and γ

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^* \cdot \vec{w}^{(t-1)}$ and γ
 - Let $(\vec{x}_i, y_i)$ be the mistake made at iteration t

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^* \cdot \vec{w}^{(t-1)}$ and γ
 - Let $(\vec{x}_i, y_i)$ be the mistake made at iteration t
 - $\vec{w}^* \cdot \vec{w}^{(t)} = \vec{w}^* \cdot (\vec{w}^{(t-1)} + y_i \vec{x}_i)$

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^* \cdot \vec{w}^{(t-1)}$ and γ
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 - $\vec{w}^* \cdot \vec{w}^{(t)} = \vec{w}^* \cdot (\vec{w}^{(t-1)} + y_i \vec{x}_i)$
 - *Definition of perceptron update*

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

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 - *Definition of perceptron update*
$$= \vec{w}^* \cdot \vec{w}^{(t-1)} + y_i \vec{w}^* \cdot \vec{x}_i$$

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
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 - *Definition of perceptron update*
$$= \vec{w}^* \cdot \vec{w}^{(t-1)} + y_i \vec{w}^* \cdot \vec{x}_i$$
 - *Distributive property of dot products, commutative scalars*

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
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$$= \vec{w}^* \cdot \vec{w}^{(t-1)} + y_i \vec{w}^* \cdot \vec{x}_i$$
 - *Distributive property of dot products, commutative scalars*
$$\geq \vec{w}^* \cdot \vec{w}^{(t-1)} + \gamma$$

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
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$$\geq \vec{w}^* \cdot \vec{w}^{(t-1)} + \gamma$$
 - *Definition of $\vec{w}^*$ and margin*

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^* \cdot \vec{w}^{(t-1)}$ and γ
 - Let $(\vec{x}_i, y_i)$ be the mistake made at iteration t
 - $\vec{w}^* \cdot \vec{w}^{(t)} = \vec{w}^* \cdot (\vec{w}^{(t-1)} + y_i \vec{x}_i)$
 - *Definition of perceptron update*
$$= \vec{w}^* \cdot \vec{w}^{(t-1)} + y_i \vec{w}^* \cdot \vec{x}_i$$
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$$\geq \vec{w}^* \cdot \vec{w}^{(t-1)} + \gamma$$
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- Then, recurse to bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

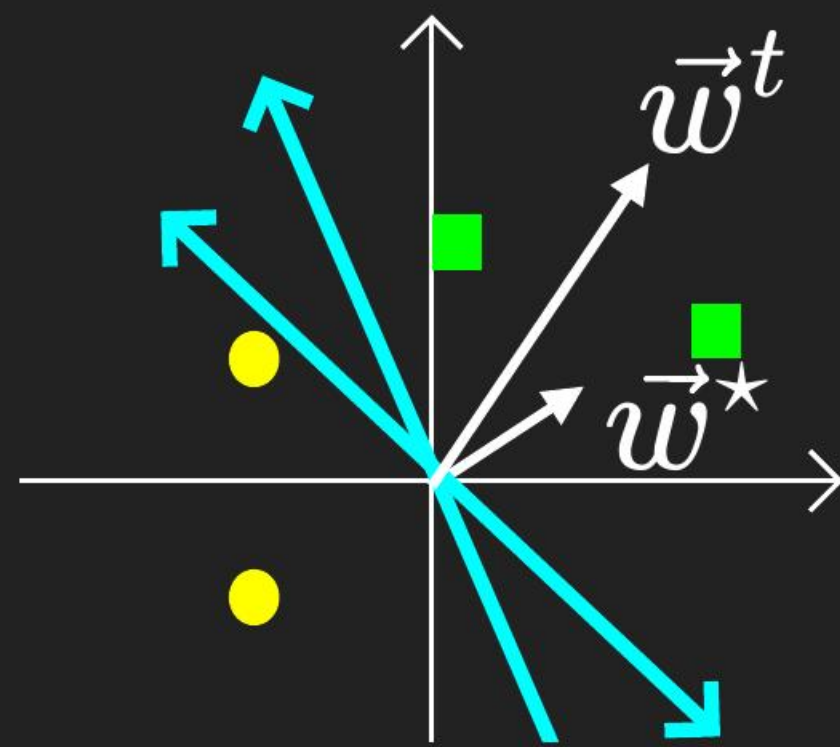
- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^* \cdot \vec{w}^{(t-1)}$ and γ
 - $\vec{w}^* \cdot \vec{w}^{(t)} \geq \vec{w}^* \cdot \vec{w}^{(t-1)} + \gamma$

Step 1: $\vec{w}^* \cdot \vec{w}^{(t)}$ is large

- Goal: lower bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
- First, bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of $\vec{w}^* \cdot \vec{w}^{(t-1)}$ and γ
 - $\vec{w}^* \cdot \vec{w}^{(t)} \geq \vec{w}^* \cdot \vec{w}^{(t-1)} + \gamma$
- Then, recurse to bound $\vec{w}^* \cdot \vec{w}^{(t)}$ in terms of γ and t
 - $\vec{w}^* \cdot \vec{w}^{(t)} \geq \vec{w}^* \cdot \vec{w}^{(t-2)} + 2\gamma$
 - $\geq \vec{w}^* \cdot \vec{w}^{(t-3)} + 3\gamma$
 - ...
 - $\geq \vec{w}^* \cdot \vec{w}^{(0)} + t\gamma$

Proof Outline

- Goal: Prove that Perceptron terminates after at most R^2/γ^2 steps
- ✓ • Step 1: $\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$
- Step 2: Show that $\|\vec{w}^{(t)}\|$ is not too large (find an upper bound)
- Step 3: combine 1 & 2 to show that if there are more than R^2/γ^2 updates, then the cosine becomes larger than one!
 - Proof by contradiction



$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$

Step 2: $\|\vec{w}^{(t)}\|$ is not too large

- Goal: upper bound $\|\vec{w}^{(t)}\|$ in terms of R and t
- First, bound $\|\vec{w}^{(t)}\|^2$ in terms of $\|\vec{w}^{(t-1)}\|^2$ and R

Step 2: $\|\vec{w}^{(t)}\|$ is not too large

- Goal: upper bound $\|\vec{w}^{(t)}\|$ in terms of R and t
- First, bound $\|\vec{w}^{(t)}\|^2$ in terms of $\|\vec{w}^{(t-1)}\|^2$ and R

Let $(\vec{x}_i, y_i)$ be the mistake made at iteration t

$$1. \|\vec{w}^{(t)}\|^2 = \|\vec{w}^{(t-1)} + y_i \vec{x}_i\|^2$$

$$2. = (\vec{w}^{(t-1)} + y_i \vec{x}_i) \cdot (\vec{w}^{(t-1)} + y_i \vec{x}_i)$$

$$3. = \vec{w}^{(t-1)} \cdot \vec{w}^{(t-1)} + 2y_i(\vec{w}^{(t-1)} \cdot \vec{x}_i) + (y_i)^2 \vec{x}_i \cdot \vec{x}_i$$

$$4. \leq \|\vec{w}^{(t-1)}\|^2 + 0 + R^2$$

1. *Definition of perceptron update*

2. *Definition of norm/dot product*

3. *Multiplying and combining terms*

4. *Mistake + radius assumption*

Step 2: $\|\vec{w}^{(t)}\|$ is not too large

- Goal: upper bound $\|\vec{w}^{(t)}\|$ in terms of R and t
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Let $(\vec{x}_i, y_i)$ be the mistake made at iteration t

$$1. \|\vec{w}^{(t)}\|^2 = \|\vec{w}^{(t-1)} + y_i \vec{x}_i\|^2$$

$$2. = (\vec{w}^{(t-1)} + y_i \vec{x}_i) \cdot (\vec{w}^{(t-1)} + y_i \vec{x}_i)$$

$$3. = \vec{w}^{(t-1)} \cdot \vec{w}^{(t-1)} + 2y_i(\vec{w}^{(t-1)} \cdot \vec{x}_i) + (y_i)^2 \vec{x}_i \cdot \vec{x}_i$$

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1. *Definition of perceptron update*

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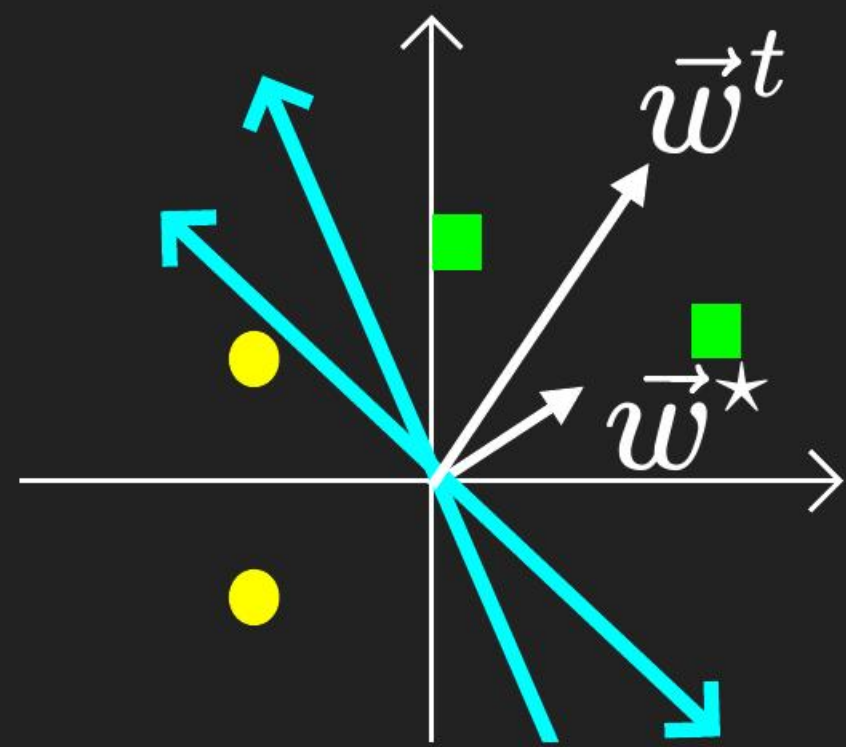
- Then, recurse to bound $\|\vec{w}^{(t)}\|^2$ in terms of R and t

Step 2: $\|\vec{w}^{(t)}\|$ is not too large

- Goal: upper bound $\|\vec{w}^{(t)}\|$ in terms of R and t
- First, bound $\|\vec{w}^{(t)}\|^2$ in terms of $\|\vec{w}^{(t-1)}\|^2$ and R
 - $\|\vec{w}^{(t)}\|^2 \leq \|\vec{w}^{(t-1)}\|^2 + R^2$
- Then, recurse to bound $\|\vec{w}^{(t)}\|^2$ in terms of R and t
 - $\|\vec{w}^{(t)}\|^2 \leq \|\vec{w}^{(t-2)}\|^2 + 2R^2$
 - ...
 - $\|\vec{w}^{(t)}\|^2 \leq \|\vec{w}^{(0)}\|^2 + tR^2$

Proof Outline

- Goal: Prove that Perceptron terminates after at most R^2/γ^2 steps
- ✓ • Step 1: $\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$
- ✓ • Step 2: $\|\vec{w}^{(t)}\|^2 \leq tR^2$
- Step 3: combine 1 & 2 to show that if there are more than R^2/γ^2 updates, then the cosine becomes larger than one!
 - Proof by contradiction



$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$

Step 3: Contradiction

- Goal: show that if $t > R^2/\gamma^2$, there is a contradiction
- We will use step 1 ($\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$) & step 2 ($\|\vec{w}^{(t)}\|^2 \leq tR^2$)

Step 3: Contradiction

- Goal: show that if $t > R^2/\gamma^2$, there is a contradiction
- We will use step 1 ($\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$) & step 2 ($\|\vec{w}^{(t)}\|^2 \leq tR^2$)
- Starting with our expression for cosine,

$$\begin{aligned}\blacksquare \cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|} \\ &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\sqrt{tR^2}} \\ &\geq \frac{\sqrt{t}\gamma}{R}\end{aligned}$$

Step 3: Contradiction

- Goal: show that if $t > R^2/\gamma^2$, there is a contradiction
- We will use step 1 ($\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$) & step 2 ($\|\vec{w}^{(t)}\|^2 \leq tR^2$)
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- *Definition of $\vec{w}^*$*

Step 3: Contradiction

- Goal: show that if $t > R^2/\gamma^2$, there is a contradiction
- We will use step 1 ($\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$) & step 2 ($\|\vec{w}^{(t)}\|^2 \leq tR^2$)
- Starting with our expression for cosine,

$$\begin{aligned}\blacksquare \cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|} \\ &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\sqrt{tR^2}} \\ &\geq \frac{\sqrt{t}\gamma}{R}\end{aligned}$$

- *Definition of $\vec{w}^*$*
- *Step 1*

Step 3: Contradiction

- Goal: show that if $t > R^2/\gamma^2$, there is a contradiction
- We will use step 1 ($\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$) & step 2 ($\|\vec{w}^{(t)}\|^2 \leq tR^2$)
- Starting with our expression for cosine,

$$\begin{aligned}\blacksquare \cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|} \\ &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\sqrt{tR^2}} \\ &\geq \frac{\sqrt{t}\gamma}{R}\end{aligned}$$

- *Definition of $\vec{w}^*$*
- *Step 1*
- *Step 2*

Step 3: Contradiction

- Goal: show that if $t > R^2/\gamma^2$, there is a contradiction
- We will use step 1 ($\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$) & step 2 ($\|\vec{w}^{(t)}\|^2 \leq tR^2$)
- Starting with our expression for cosine,

$$\begin{aligned}\blacksquare \cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|} \\ &= \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\|\vec{w}^{(t)}\|} \\ &\geq \frac{t\gamma}{\sqrt{tR^2}} \\ &\geq \frac{\sqrt{t}\gamma}{R}\end{aligned}$$

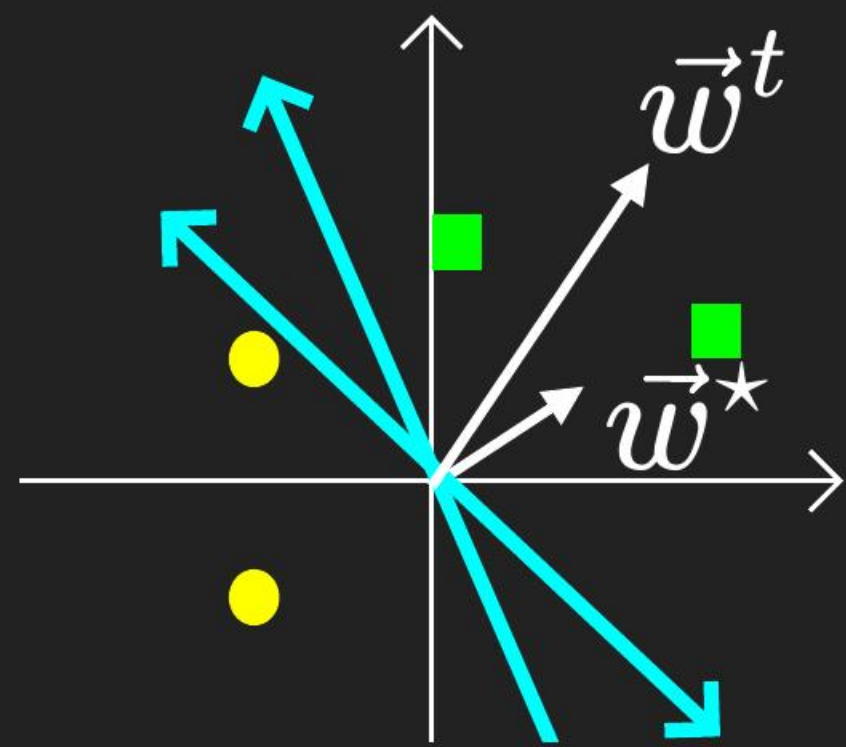
- *Definition of $\vec{w}^*$*
- *Step 1*
- *Step 2*
- *division*

- If $t > R^2/\gamma^2$,

$$\blacksquare \cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) > \frac{\sqrt{R^2/\gamma^2}\gamma}{R} = 1 \implies \text{contradiction!}$$

Proof Outline

- Goal: Prove that Perceptron terminates after at most R^2/γ^2 steps
- ✓ • Step 1: $\vec{w}^* \cdot \vec{w}^{(t)} \geq t\gamma$
- ✓ • Step 2: $\|\vec{w}^{(t)}\|^2 \leq tR^2$
- ✓ • Step 3: If $t > R^2/\gamma^2$, $\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) > 1$
 - This is impossible, so it must be that $t \leq R^2/\gamma^2$



$$\cos(\theta_{\vec{w}^*, \vec{w}^{(t)}}) = \frac{\vec{w}^* \cdot \vec{w}^{(t)}}{\|\vec{w}^*\| \|\vec{w}^{(t)}\|}$$

Remarks on Convergence

- **Theorem:** Given a dataset $\mathcal{S} = \{(\vec{x}_1, y_1), \dots, (\vec{x}_m, y_m)\}$ and a radius R such that $\|\vec{x}_i\| \leq R$ for all $i \in [m]$. If $\mathcal{S}$ is linearly separable with (geometric) **margin** γ , then Perceptron makes at most R^2/γ^2 updates before finding a consistent linear classifier.
- This upper bound holds even if
 - We scale every instance in the training set
 - We shuffle the training set
 - We don't know the value of γ
- Actual number of updates can vary, but we know it cannot be more than R^2/γ^2